Object Detection and Segmentation

Lior Yariv

February 28th, 2022

Previously

Classification

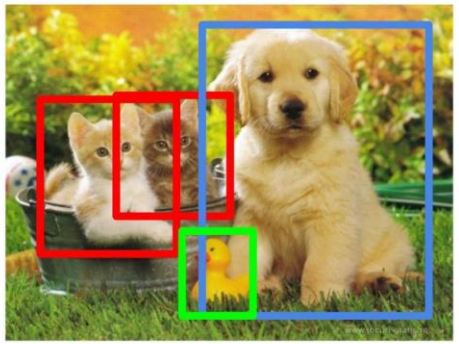
CAT

Classification + Localization



CAT

Object Detection



CAT, DOG, DUCK

Semantic Segmentation



CAT, DOG, DUCK

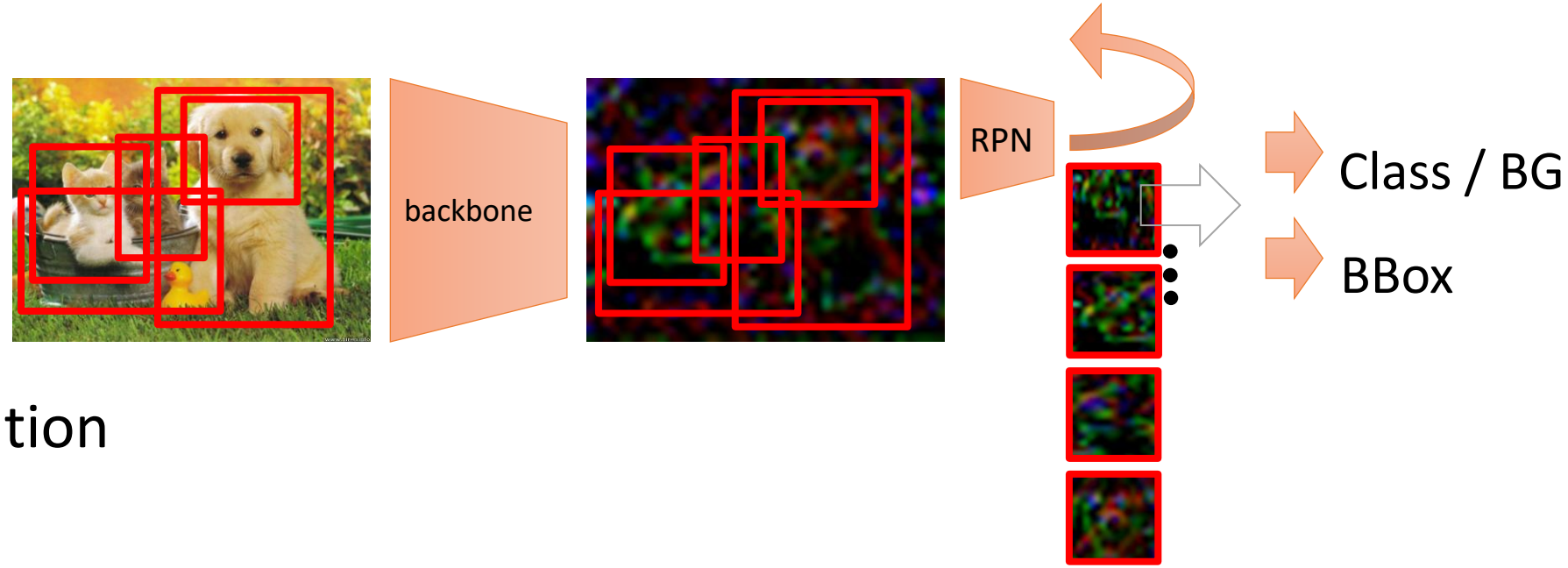
Instance Segmentation



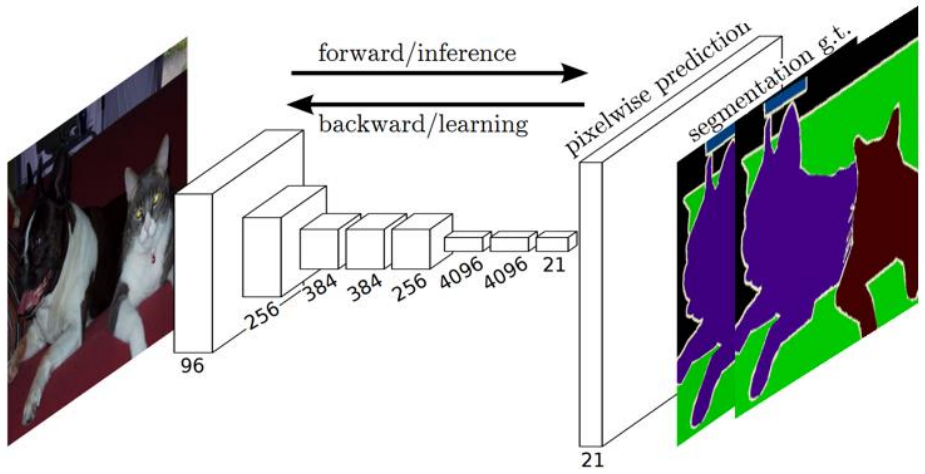
CAT, DOG, DUCK

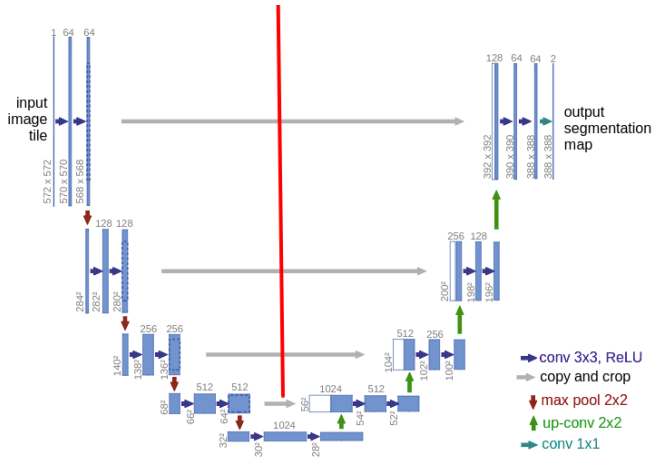
Previously

- Object detection
  - Faster RCNN



- Semantic segmentation
  - FCN / DeepLab
  - UNet





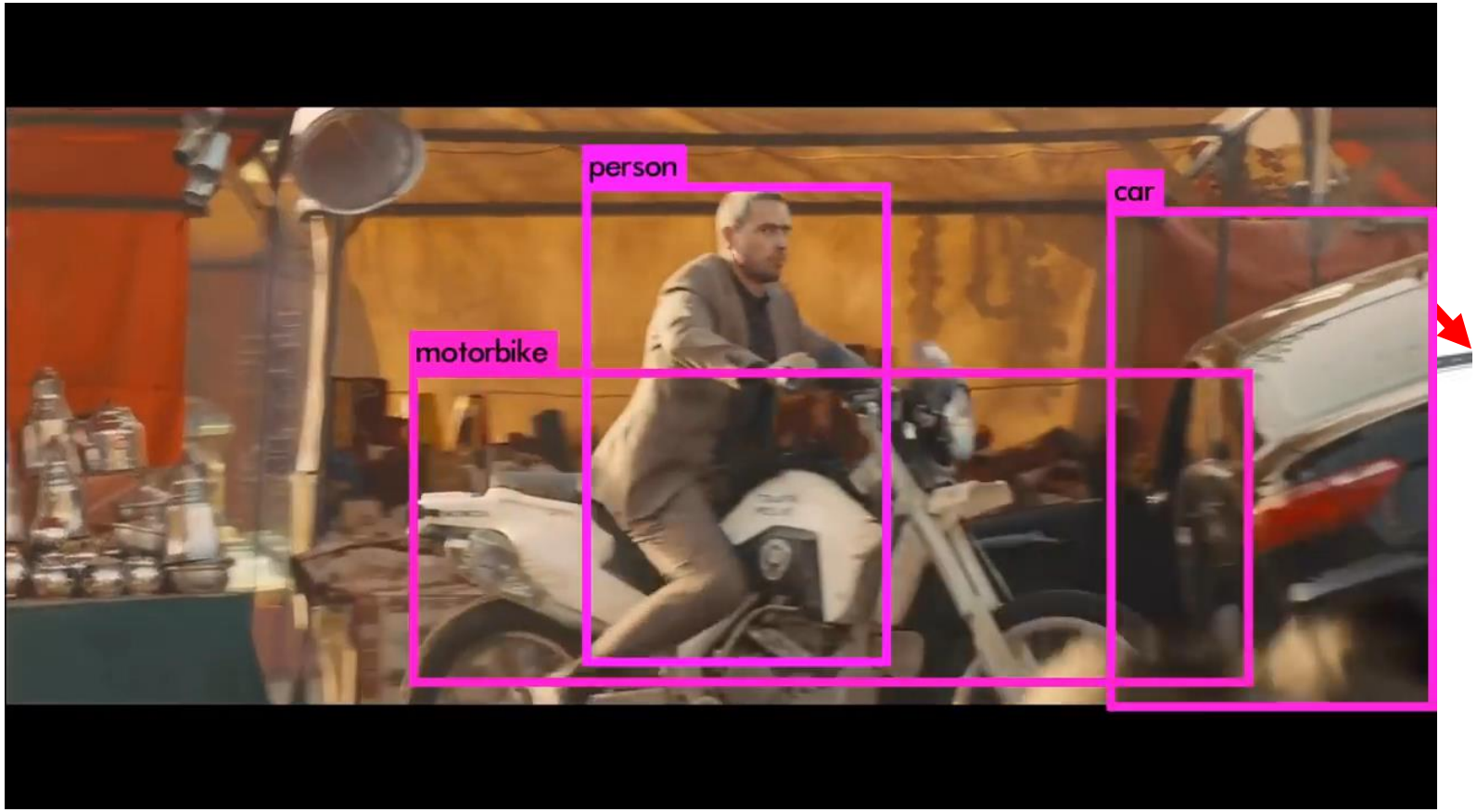
Previously

- Object detection
  - Faster RCNN
- Semantic segmentation
  - FCN / DeepLab
  - UNet

Today

- (Multiple) Object detection
  - YOLO (You Only Look Once)
  - DETR (DEtection TRansformer)
- Semantic Segmentation
  - Segmenter

Object Detection



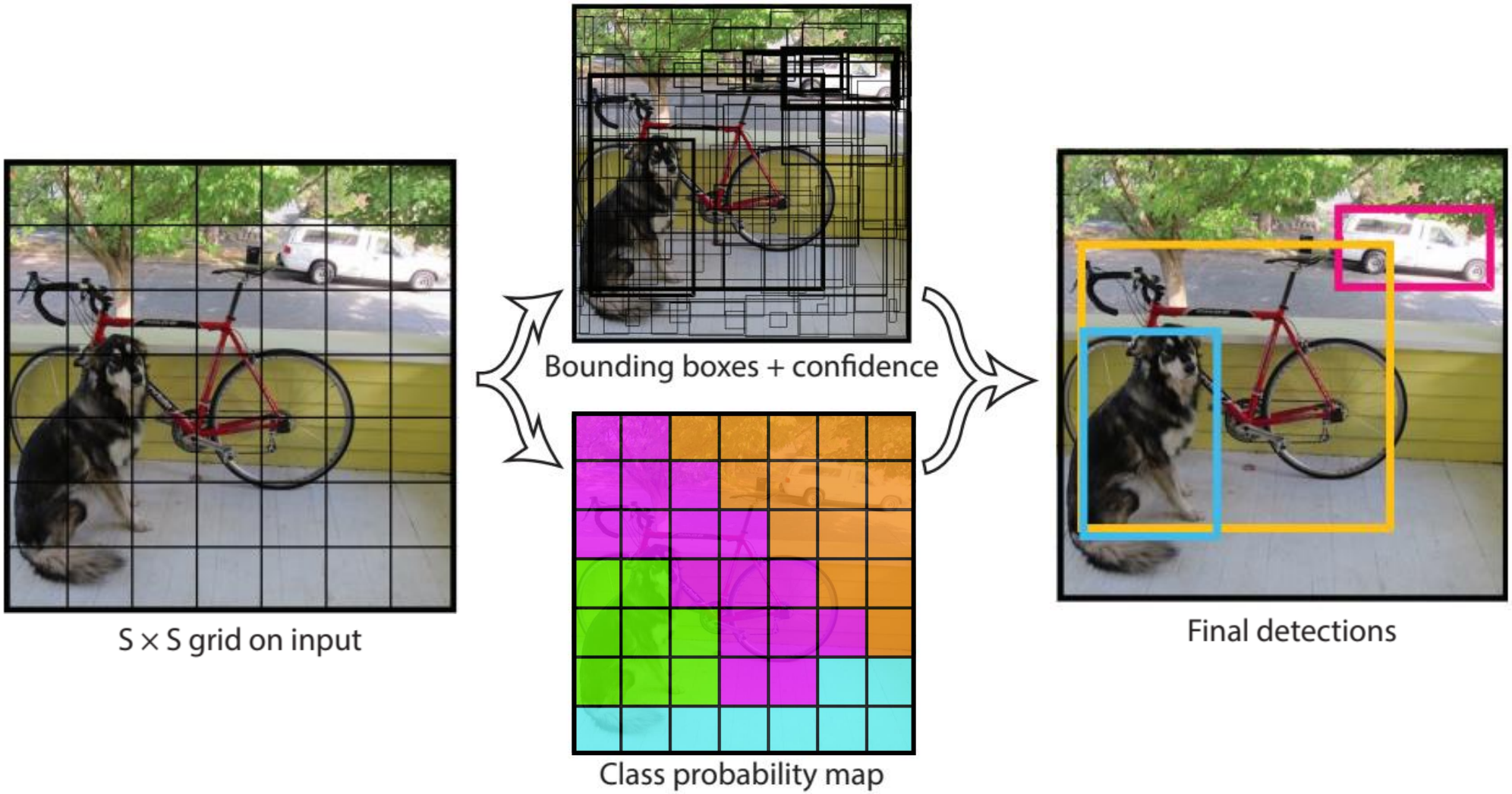
Single Shot:
SSD, YOLO …

Fast
High false rate

Video: YOLOv2

Slide credit: Shai Bagon

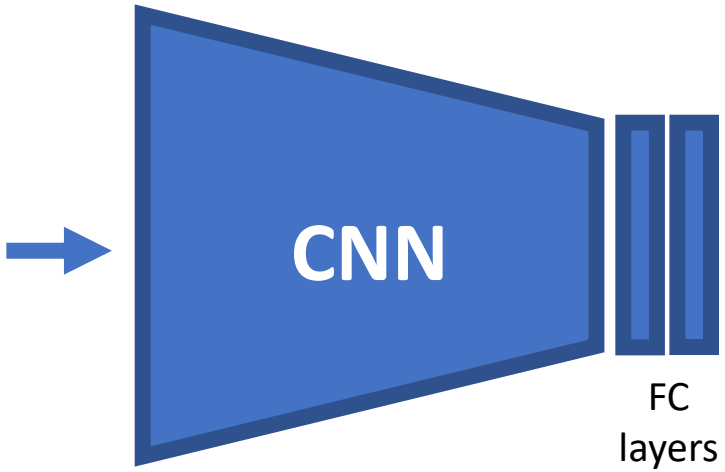
YOLO – Overview

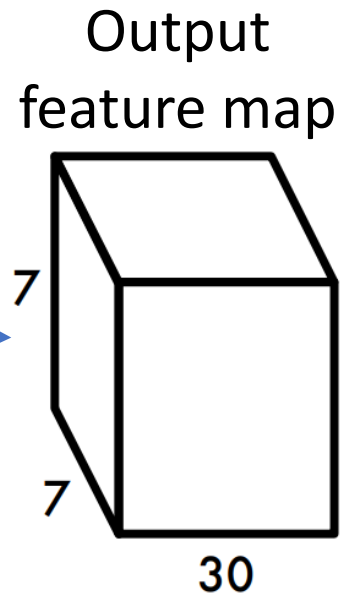


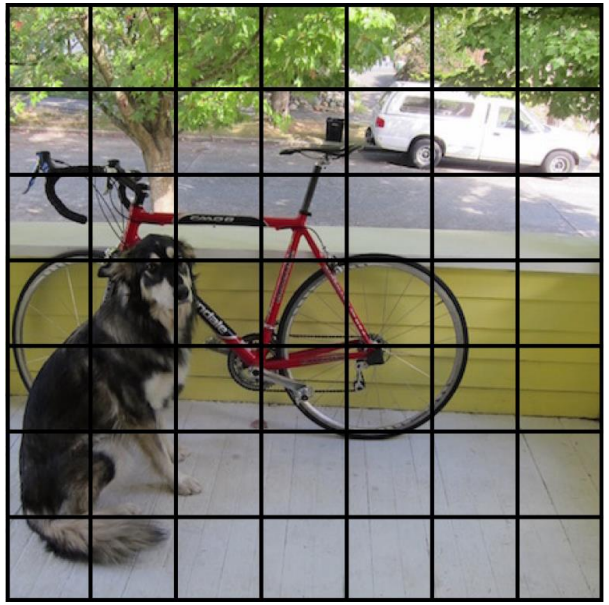
J. Redmon, S. Divvala, R. Girshick, A. Farhadi. You only look once: Unified, real-time object detection, 2015 (CVPR 2016)

YOLO









J. Redmon, S. Divvala, R. Girshick, A. Farhadi. You only look once: Unified, real-time object detection, 2015 (CVPR 2016)

YOLO

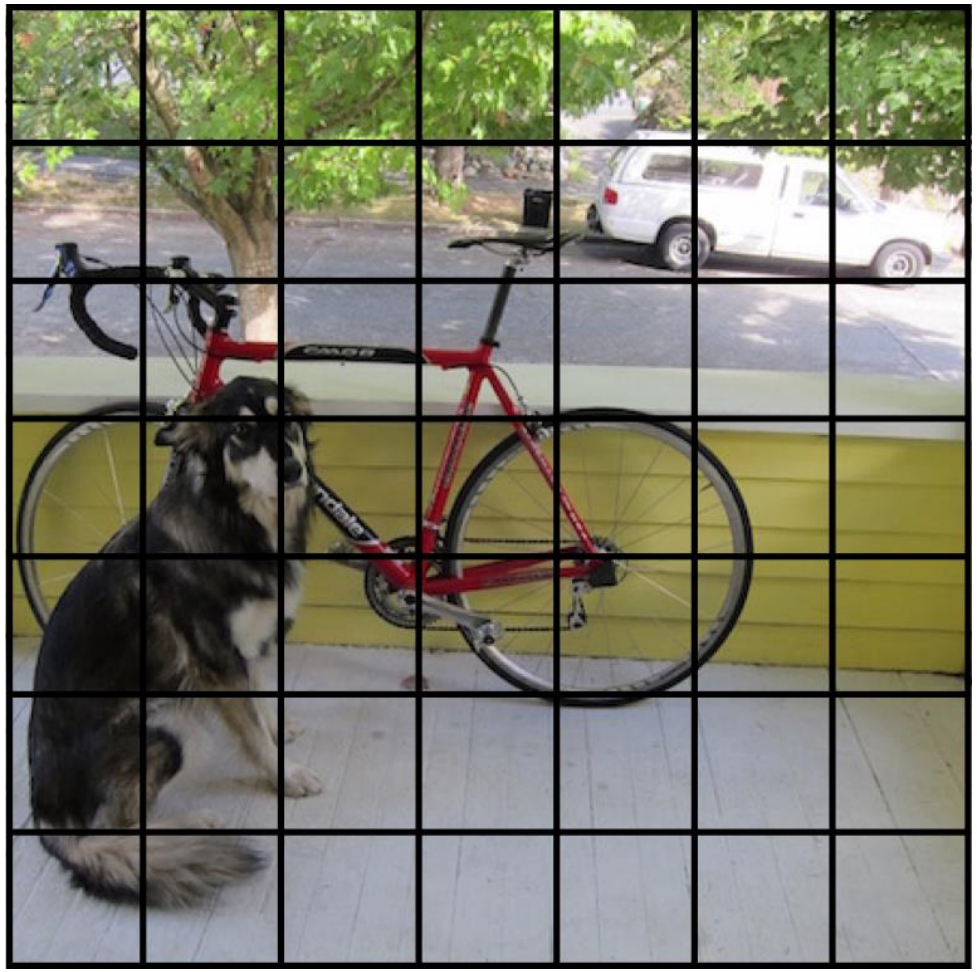


Image credit: J. Redmon, S. Divvala, R. Girshick, A. Farhadi

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YOLO

Each cell predicts

- $B = 2$ bounding boxes (x, y, w, h) + confidence score

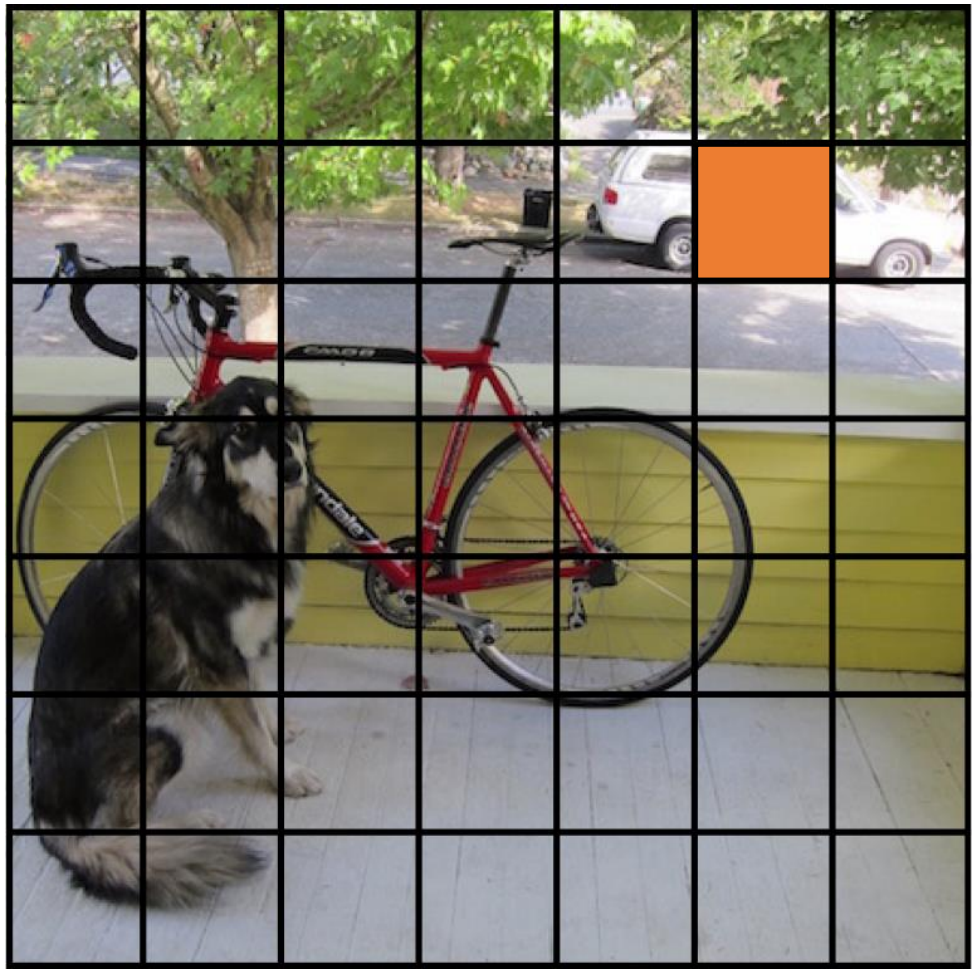


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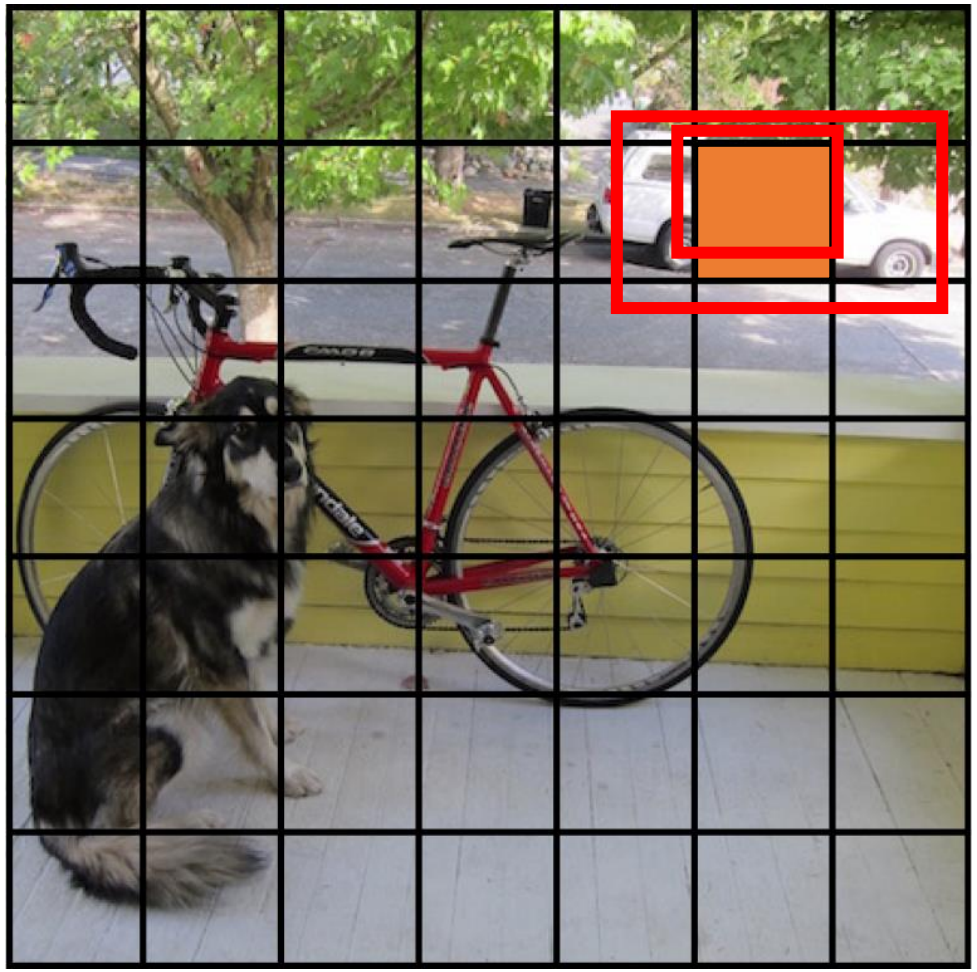


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YOLO – Inference

Each cell predicts

- $B = 2$ bounding boxes (x, y, w, h) + confidence score
- $C = 20$ class probabilities

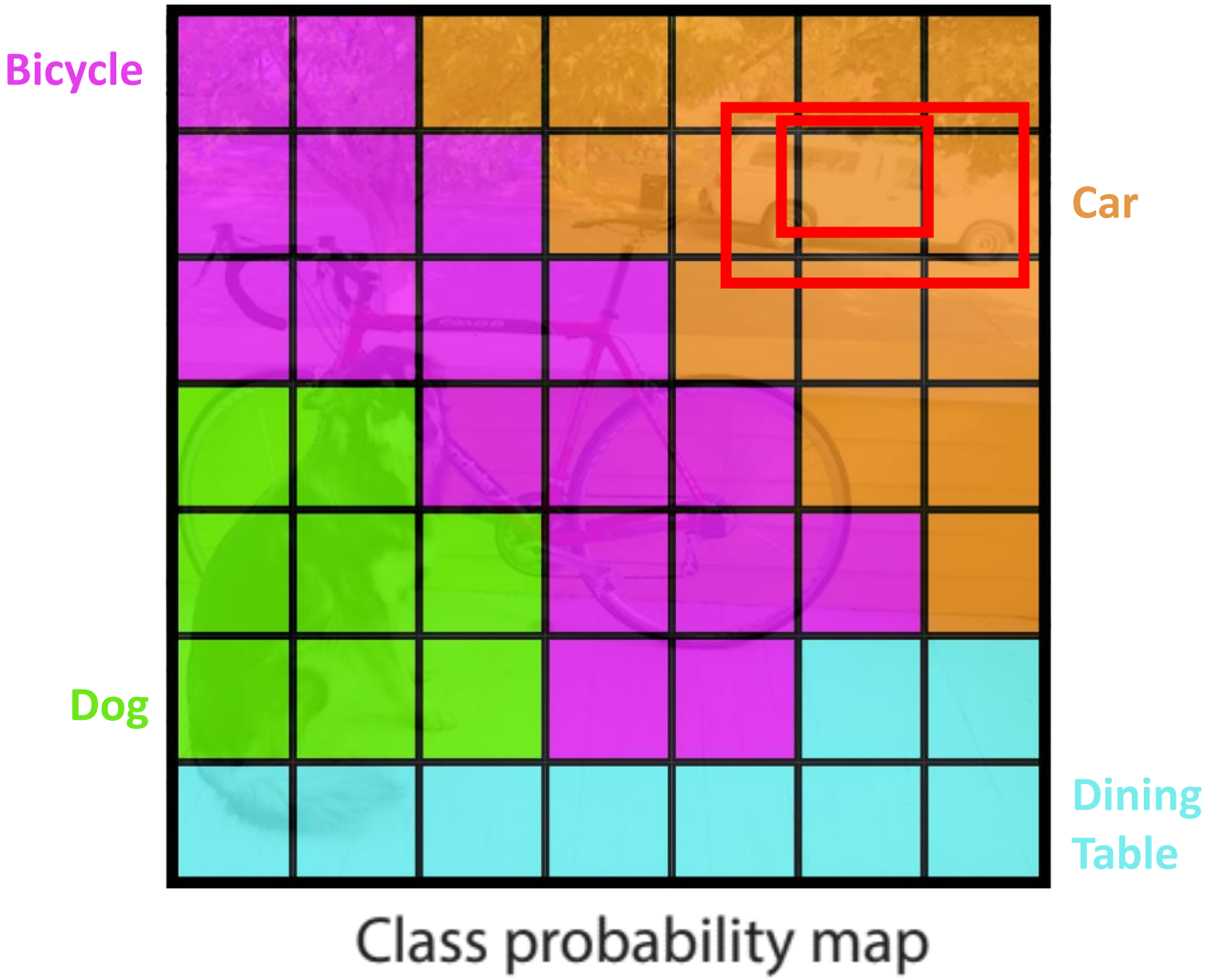
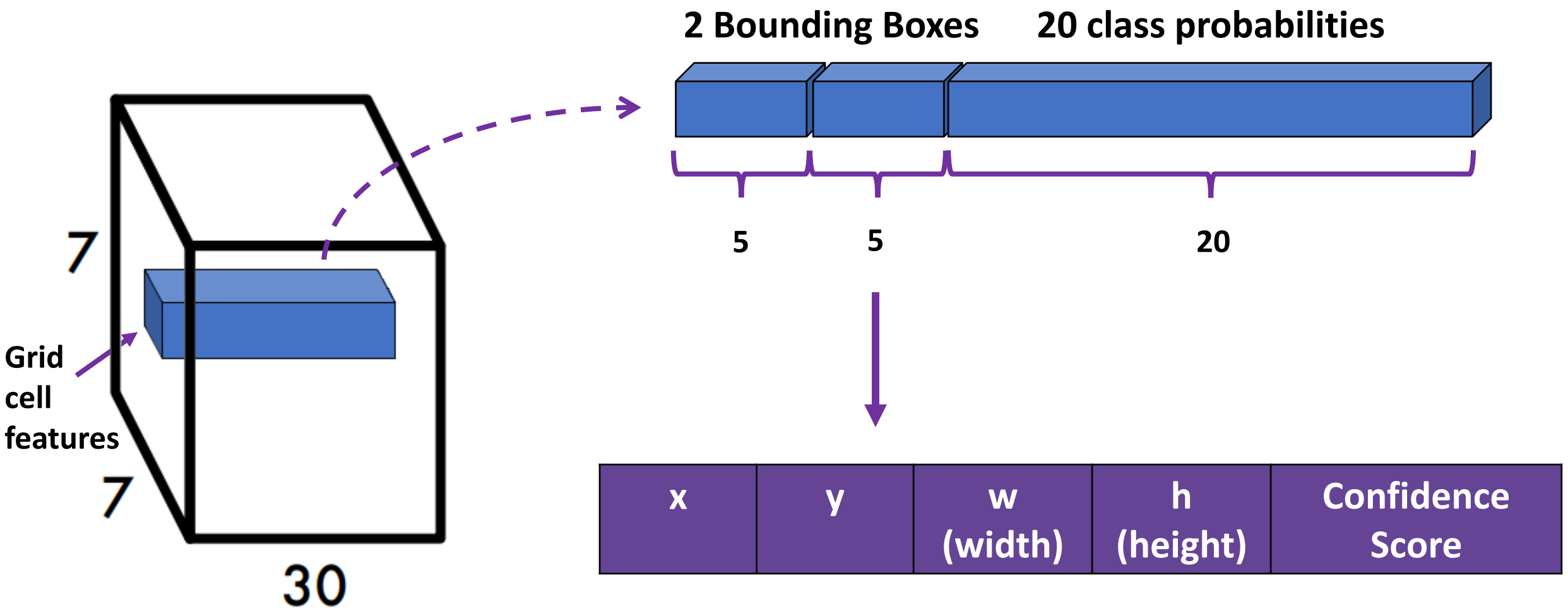


Image credit: J. Redmon, S. Divvala, R. Girshick, A. Farhadi

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YOLO – Output feature map



J. Redmon, S. Divvala, R. Girshick, A. Farhadi. You only look once: Unified, real-time object detection, 2015 (CVPR 2016)

YOLO

Each cell predicts

- $B = 2$ bounding boxes (x, y, w, h) + confidence score
- $C = 20$ class probabilities

SxSxB Bounding-Boxes (S=7,B=2 → 96 Bboxs)



S × S grid on input

Image credit: J. Redmon, S. Divvala, R. Girshick, A. Farhadi

J. Redmon, S. Divvala, R. Girshick, A. Farhadi. You only look once: Unified, real-time object detection, 2015 (CVPR 2016)

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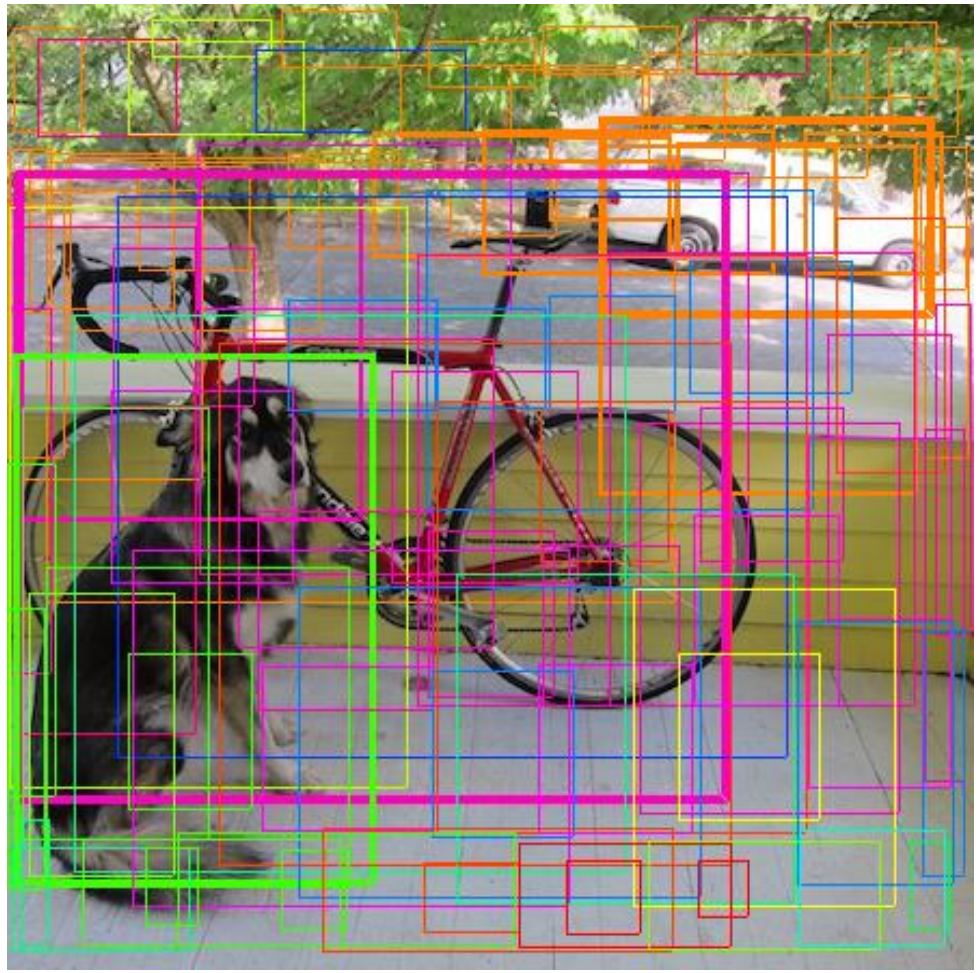


Image credit: J. Redmon, S. Divvala, R. Girshick, A. Farhadi

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YOLO

Each cell predicts

- $B = 2$ bounding boxes (x, y, w, h) + confidence score
- $C = 20$ class probabilities

- Apply Non-Maximum Suppression

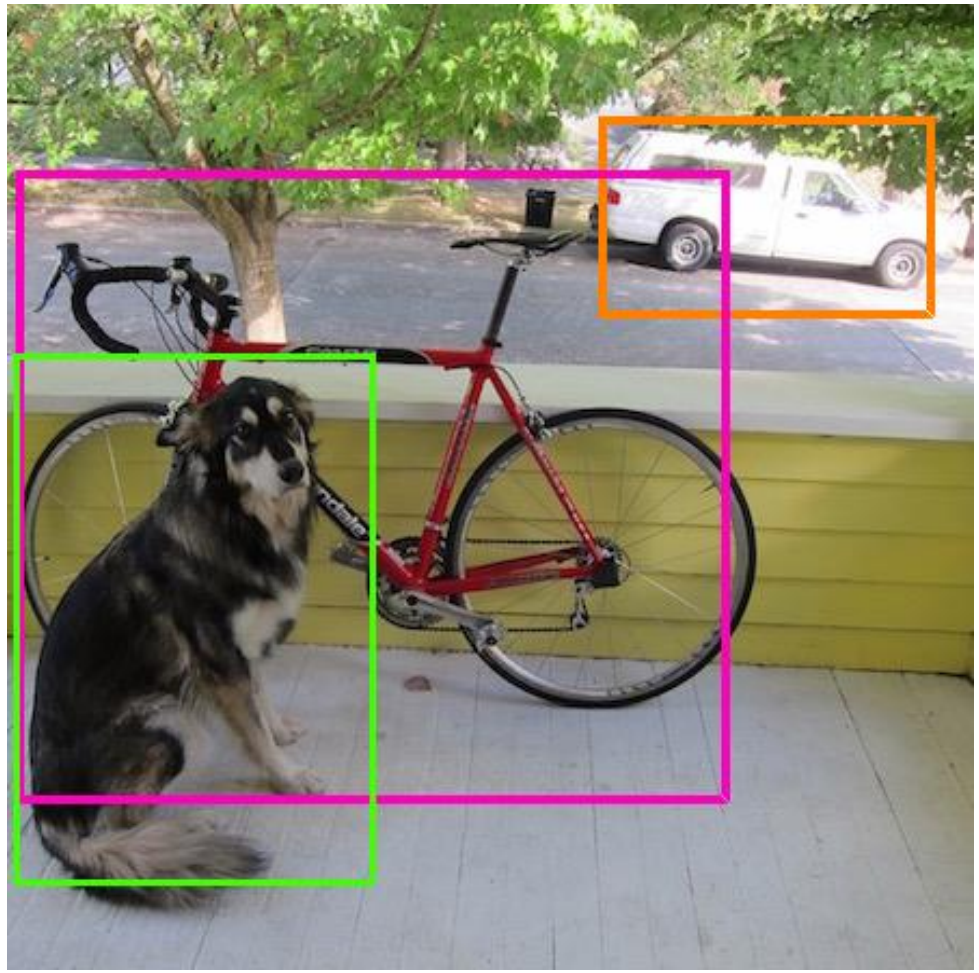


Image credit: J. Redmon, S. Divvala, R. Girshick, A. Farhadi

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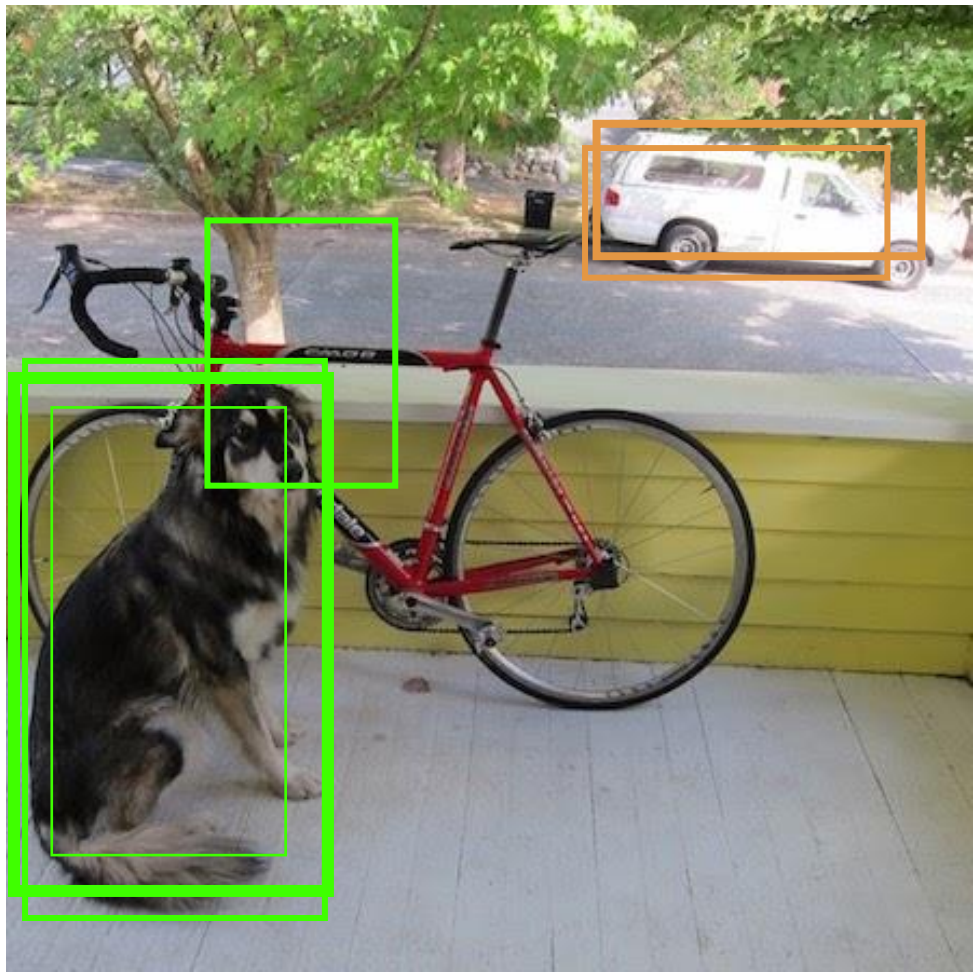
YOLO - NMS

box 1

Union

box 2

IoU = ——————

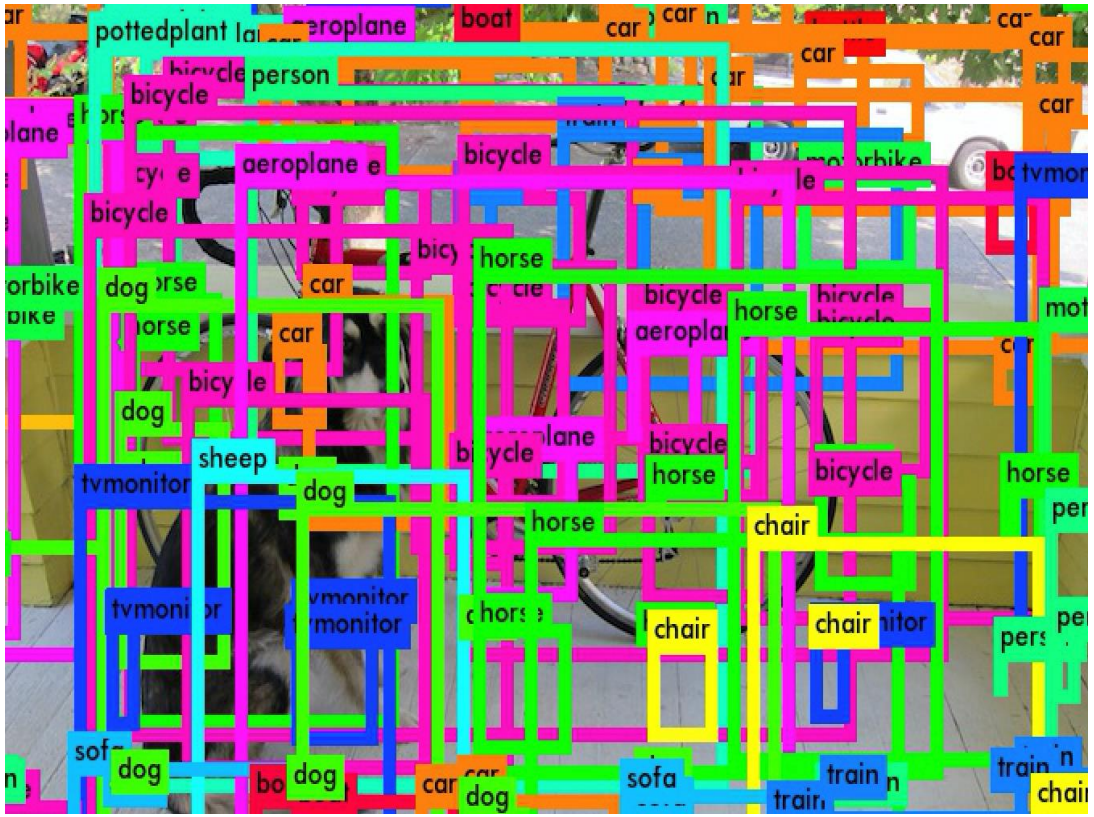


Slide idea: Shai Bagon

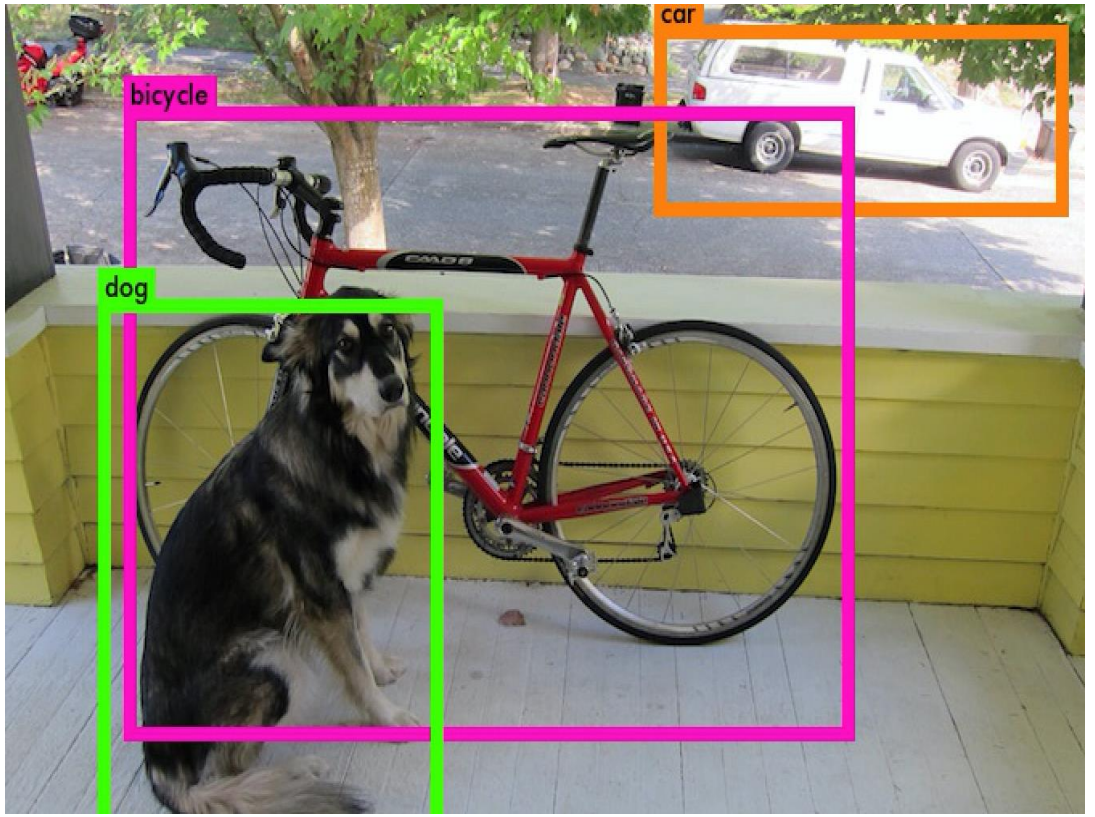
J. Redmon, S. Divvala, R. Girshick, A. Farhadi. You only look once: Unified, real-time object detection, 2015 (CVPR 2016)

YOLO – Role of Threshold

Low Threshold



High Threshold



J. Redmon, S. Divvala, R. Girshick, A. Farhadi. You only look once: Unified, real-time object detection, 2015 (CVPR 2016)

YOLO – Training (end-to-end)

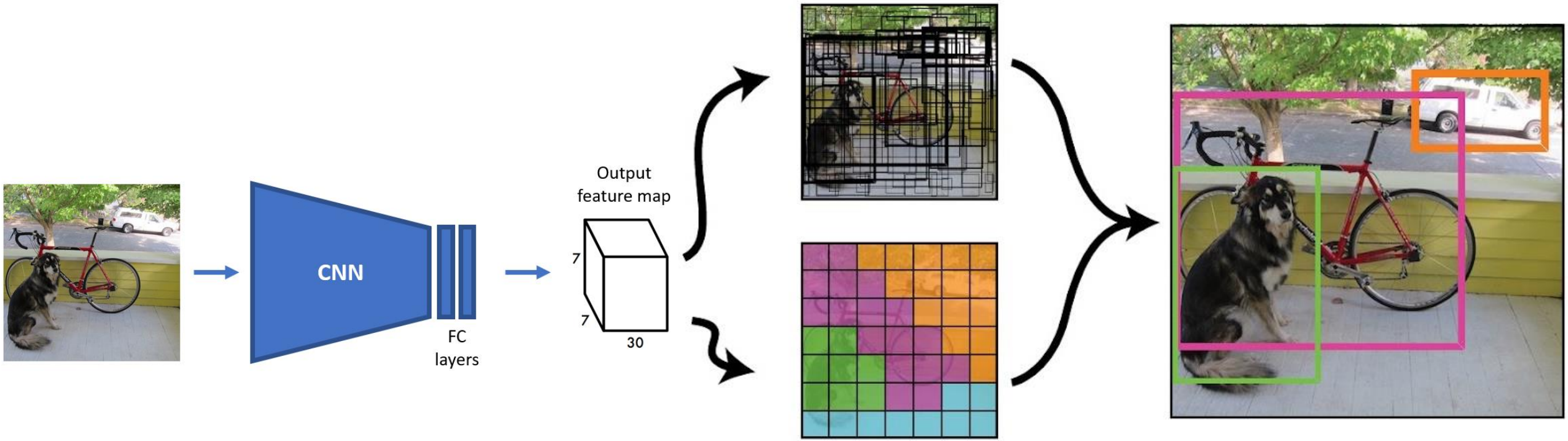


Image credit: J. Redmon, S. Divvala, R. Girshick, A. Farhadi

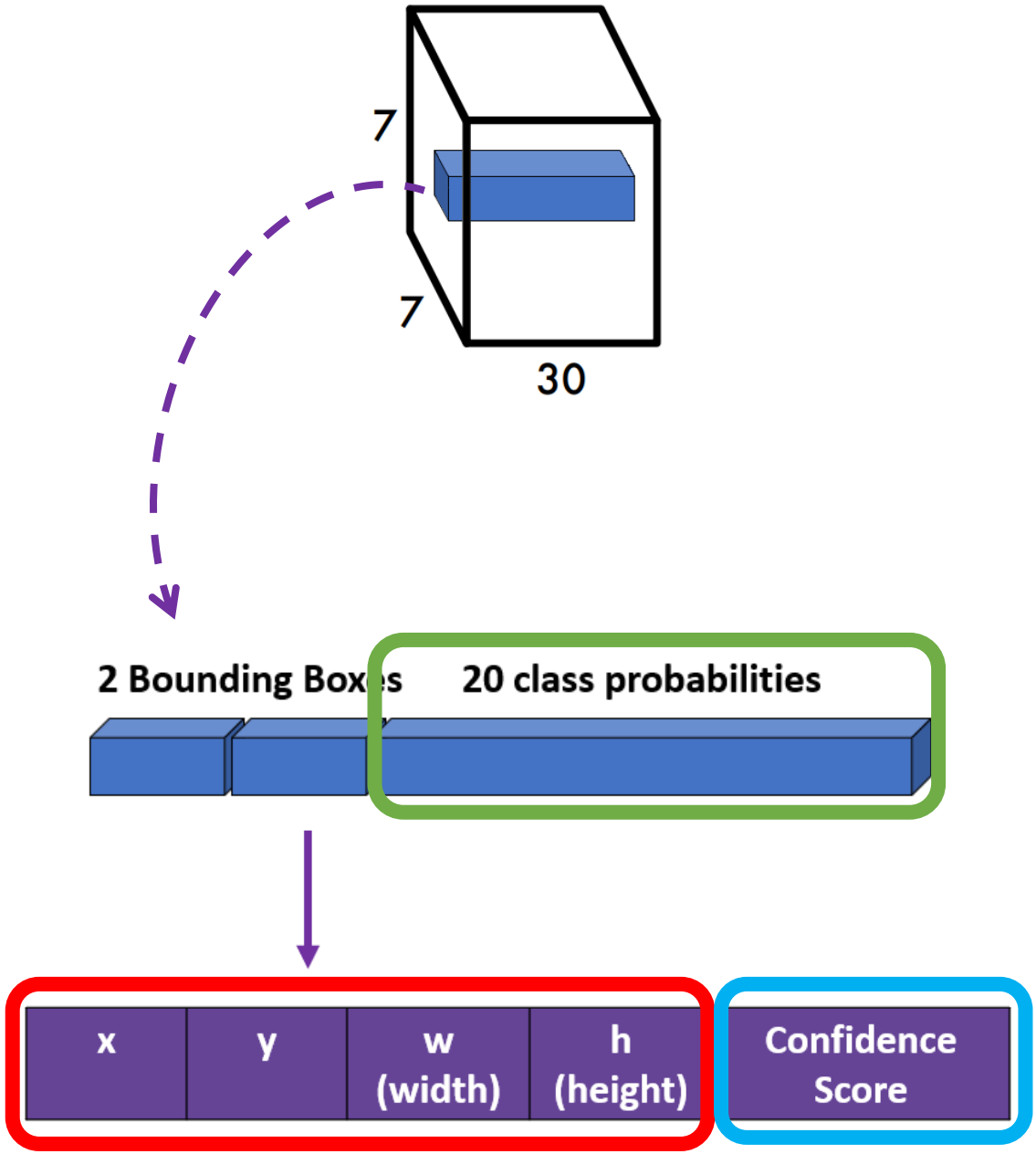
J. Redmon, S. Divvala, R. Girshick, A. Farhadi. You only look once: Unified, real-time object detection, 2015 (CVPR 2016)

YOLO – Loss function

$$\mathcal{L} = \mathcal{L}_{Localization\ Loss}$$

$$+ \mathcal{L}_{Confidence\ Loss}$$

$$+ \mathcal{L}_{Classification\ Loss}$$



J. Redmon, S. Divvala, R. Girshick, A. Farhadi. You only look once: Unified, real-time object detection, 2015 (CVPR 2016)

YOLO – Training

$$\mathcal{L} = \mathcal{L}_{Localization\ Loss} + \mathcal{L}_{Confidence\ Loss} + \mathcal{L}_{Classification\ Loss}$$

Ground truth bounding box

Center of object

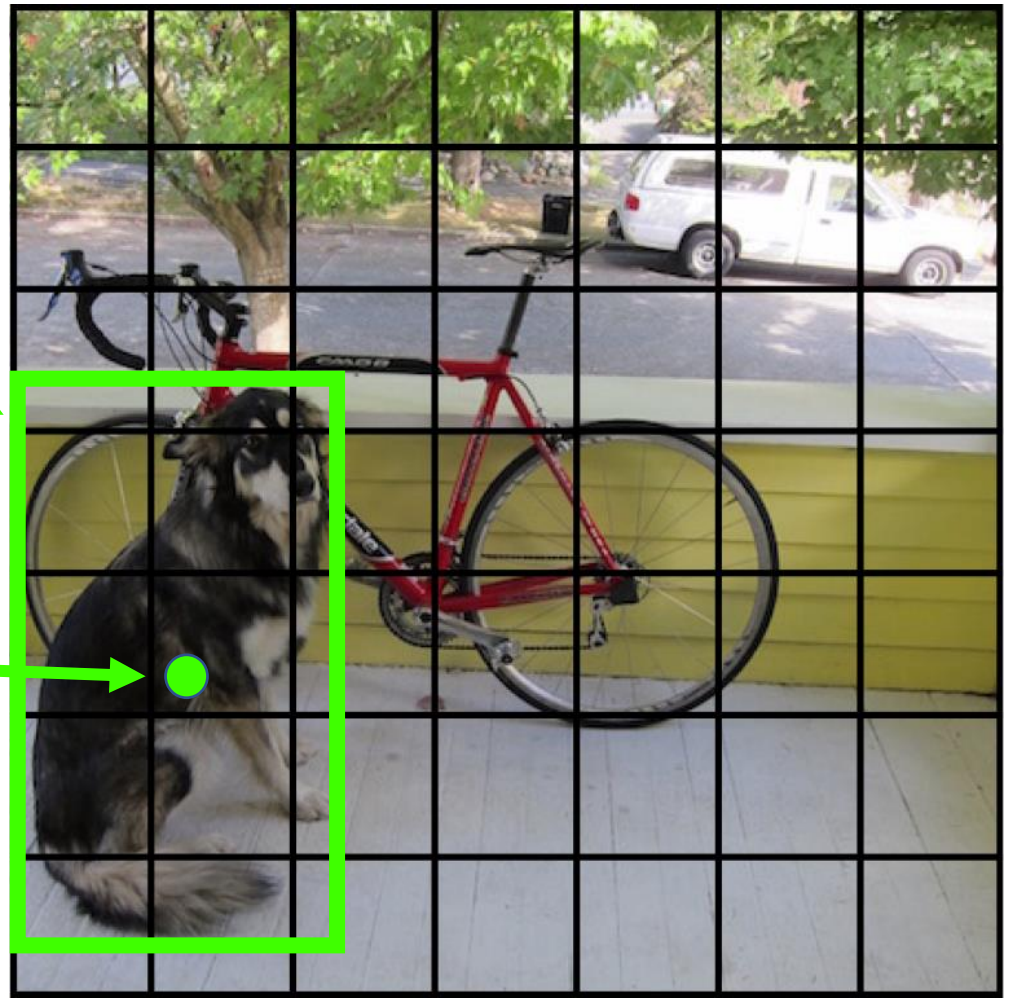


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YOLO – Training

$$\mathcal{L} = \mathcal{L}_{Localization\ Loss} + \mathcal{L}_{Confidence\ Loss} + \mathcal{L}_{Classification\ Loss}$$

Ground truth bounding box

Assign to specific cell

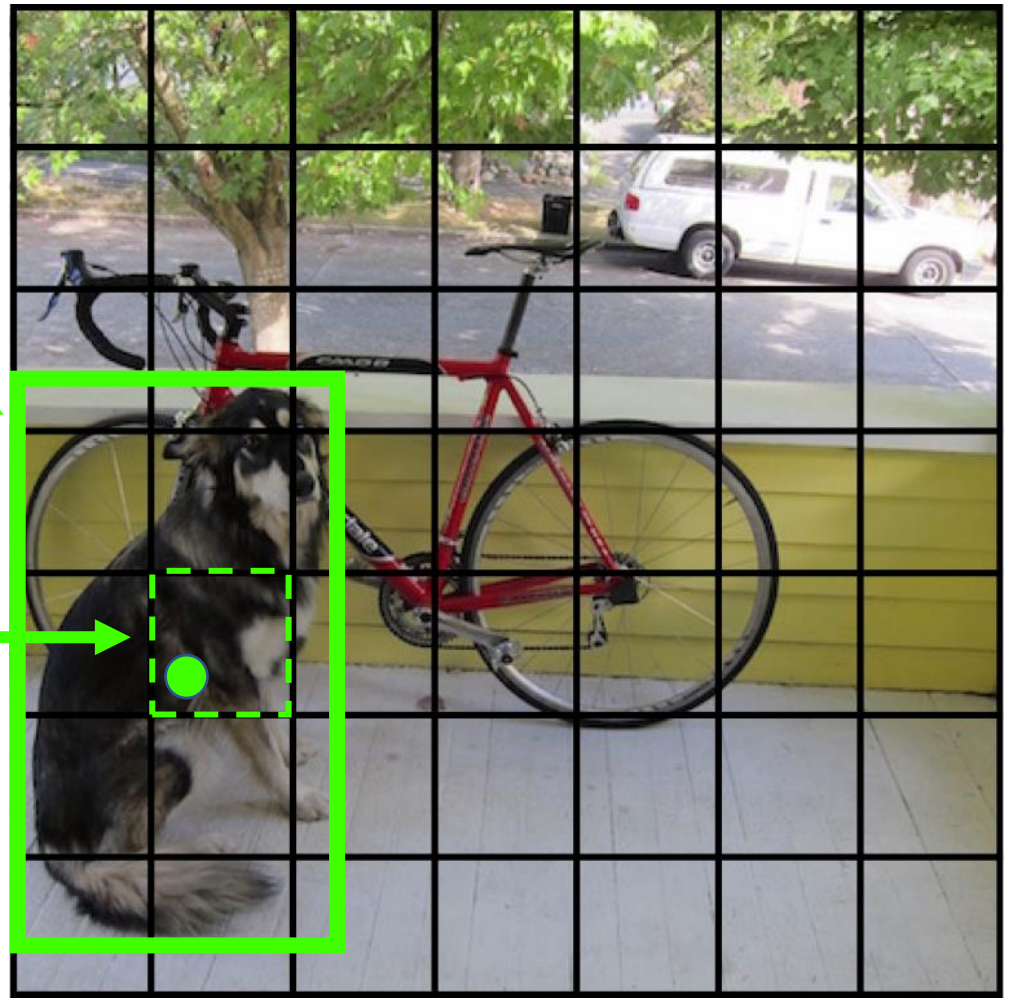


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YOLO – Training

$$\mathcal{L} = \mathcal{L}_{Localization\ Loss} + \mathcal{L}_{Confidence\ Loss} + \mathcal{L}_{Classification\ Loss}$$

Ground truth bounding box

Supervisory signal:

Dog: 1
Cat: 0
Bike: 0
…

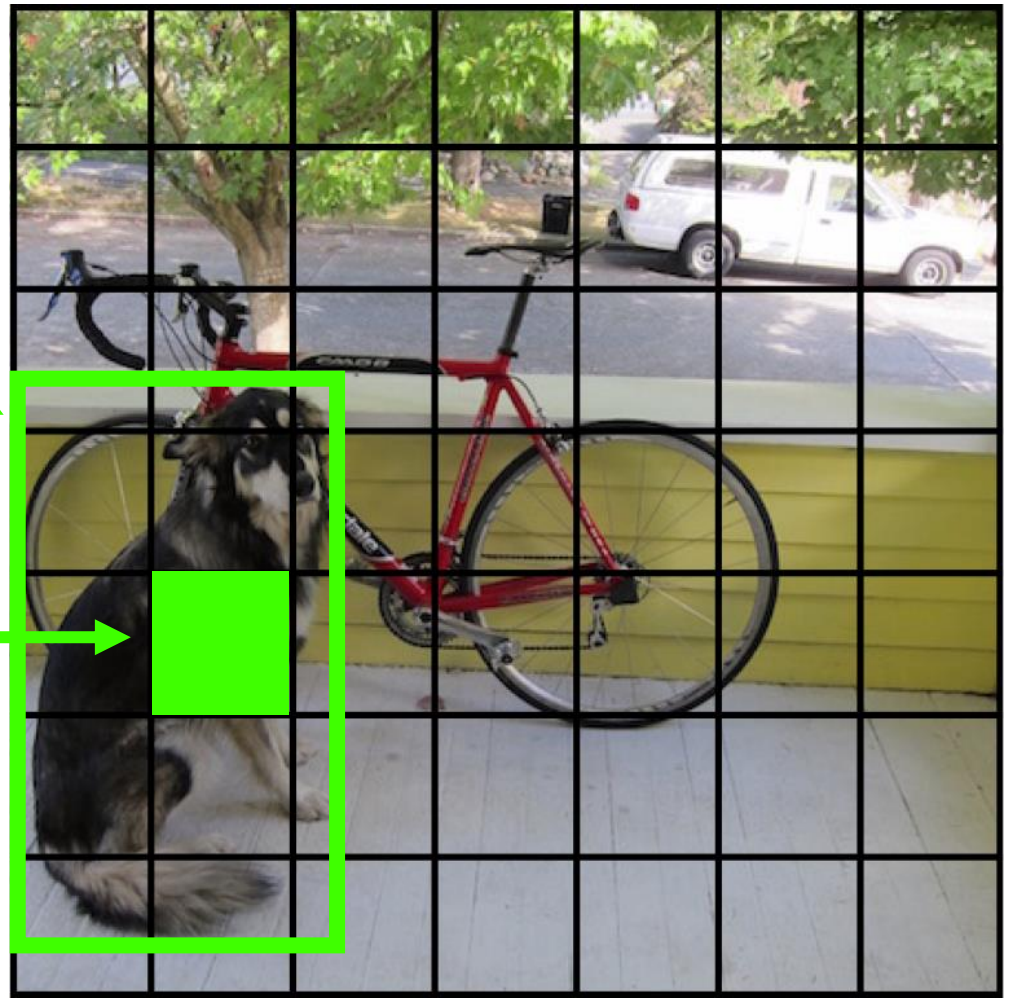


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YOLO – Training

$$\mathcal{L} = \mathcal{L}_{Localization\ Loss} + \mathcal{L}_{Confidence\ Loss} + \mathcal{L}_{Classification\ Loss}$$

Ground truth bounding box

Look at cell's predicted boxes

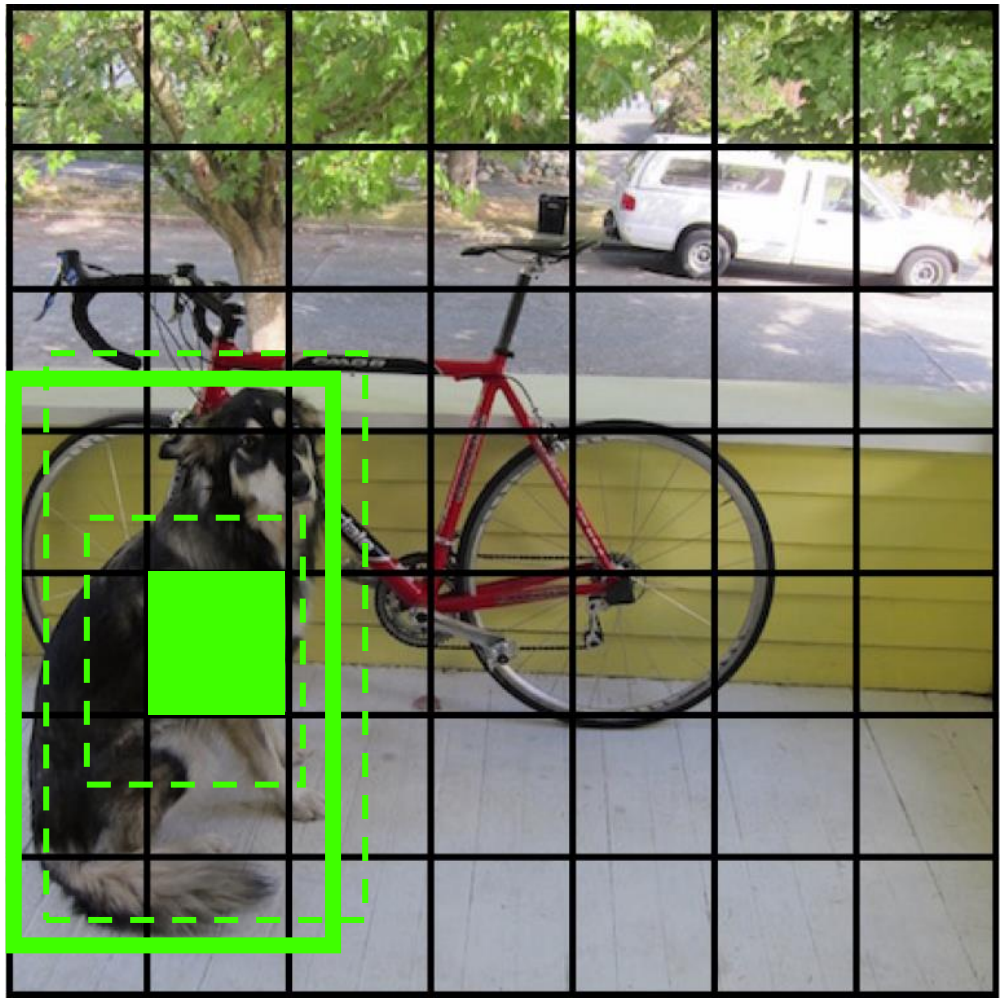


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YOLO – Training

$$\mathcal{L} = \mathcal{L}_{Localization\ Loss} + \mathcal{L}_{Confidence\ Loss} + \mathcal{L}_{Classification\ Loss}$$

Ground truth bounding box

Increase confidence score

adjust Llocalization

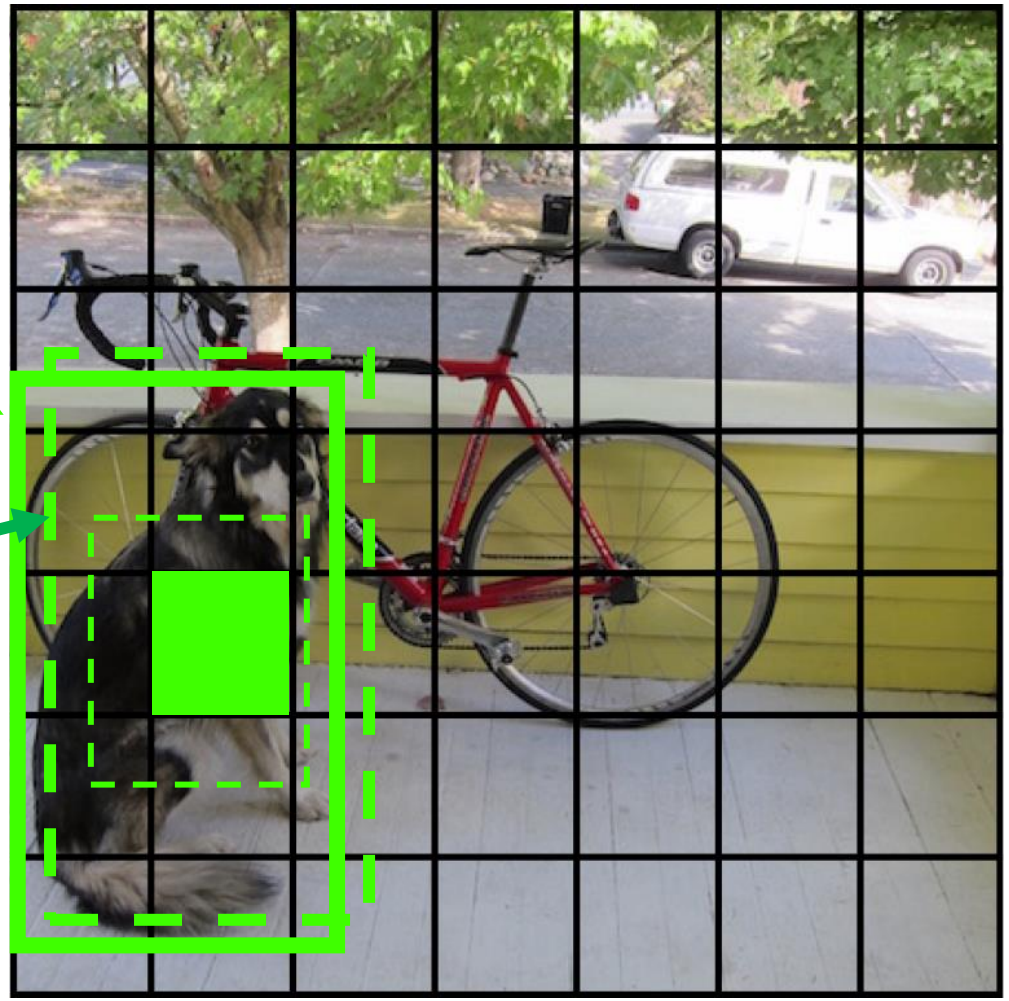


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YOLO – Training

$$\mathcal{L} = \mathcal{L}_{\text{Localization Loss}}$$

$$+ \mathcal{L}_{\text{Confidence Loss}}$$

$$+ \mathcal{L}_{\text{Classification Loss}}$$

Ground truth bounding box

Increase confidence score adjust $L_{\text{localization}}$

Decrease confidence score don't adjust $L_{\text{localization}}$

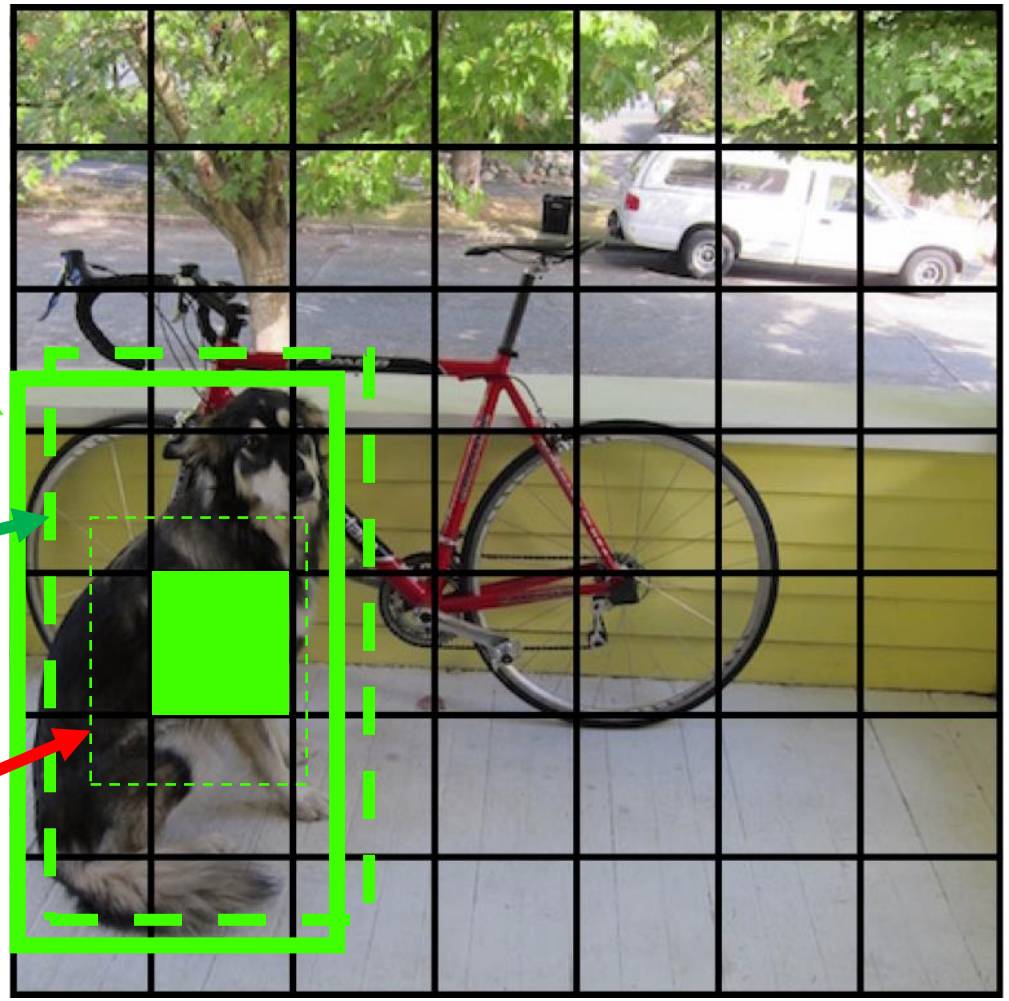


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YOLO – Training

$$\mathcal{L} = \mathcal{L}_{Localization\ Loss} + \mathcal{L}_{Confidence\ Loss} + \mathcal{L}_{Classification\ Loss}$$

A cell with no ground truth detection

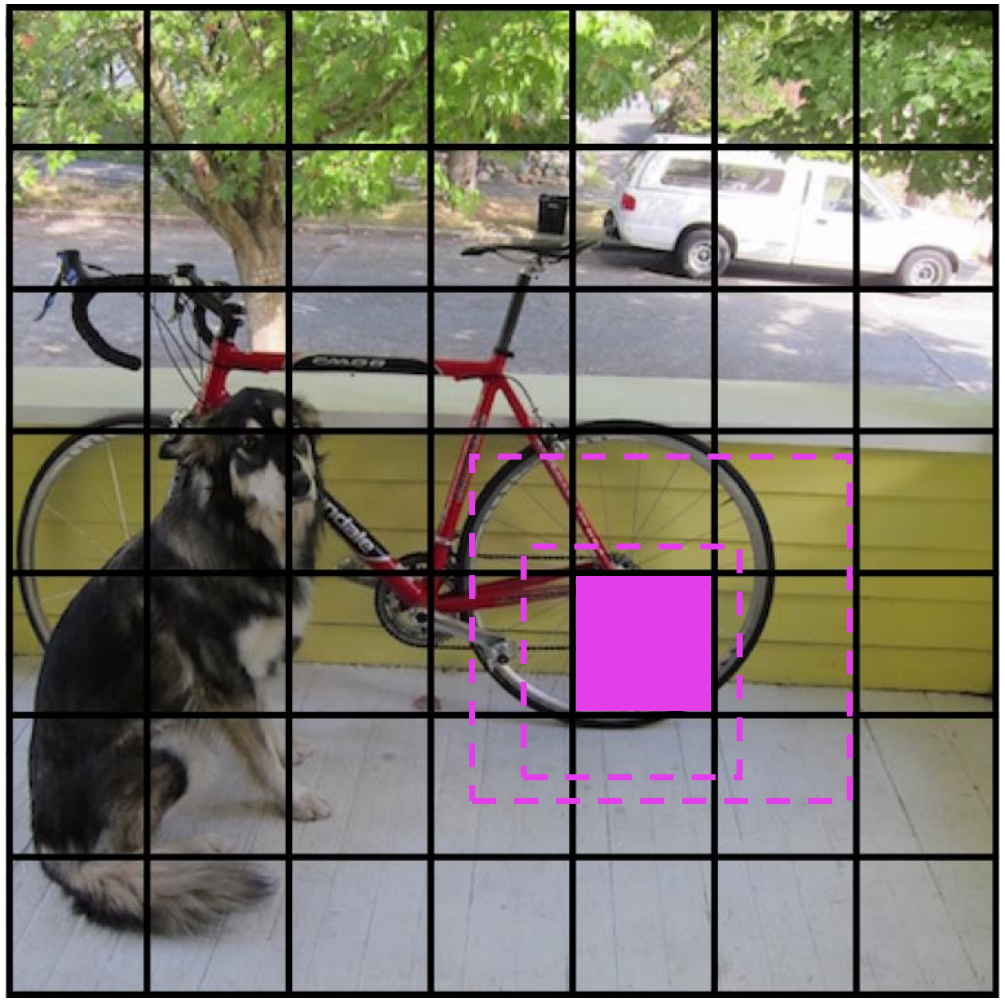


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YOLO – Training

$$\mathcal{L} = \mathcal{L}_{Localization\ Loss} + \mathcal{L}_{Confidence\ Loss} + \mathcal{L}_{Classification\ Loss}$$

A cell with no ground truth detection

Decrease confidence score

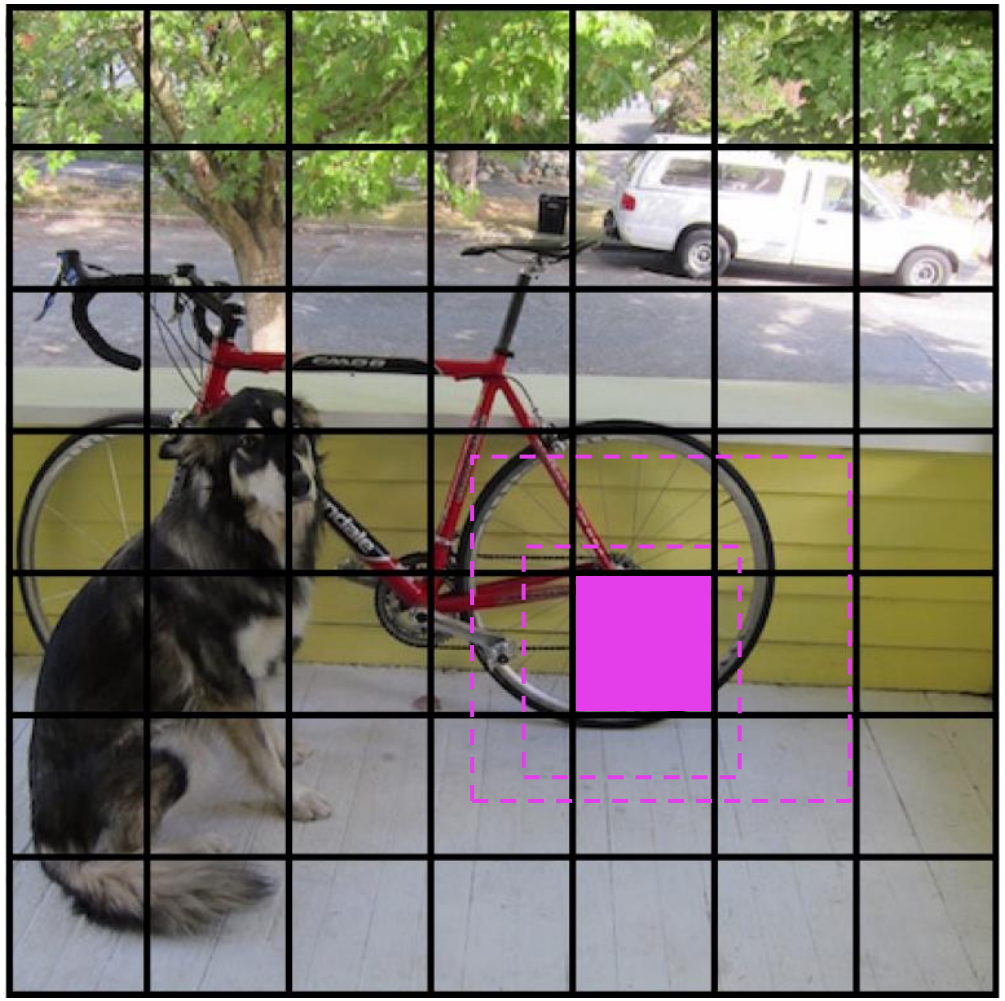


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YOLO – Training

$$\mathcal{L} = \mathcal{L}_{\text{Localization Loss}} + \mathcal{L}_{\text{Confidence Loss}} + \mathcal{L}_{\text{Classification Loss}}$$

A cell with no ground truth detection

Decrease confidence score

Don't adjust

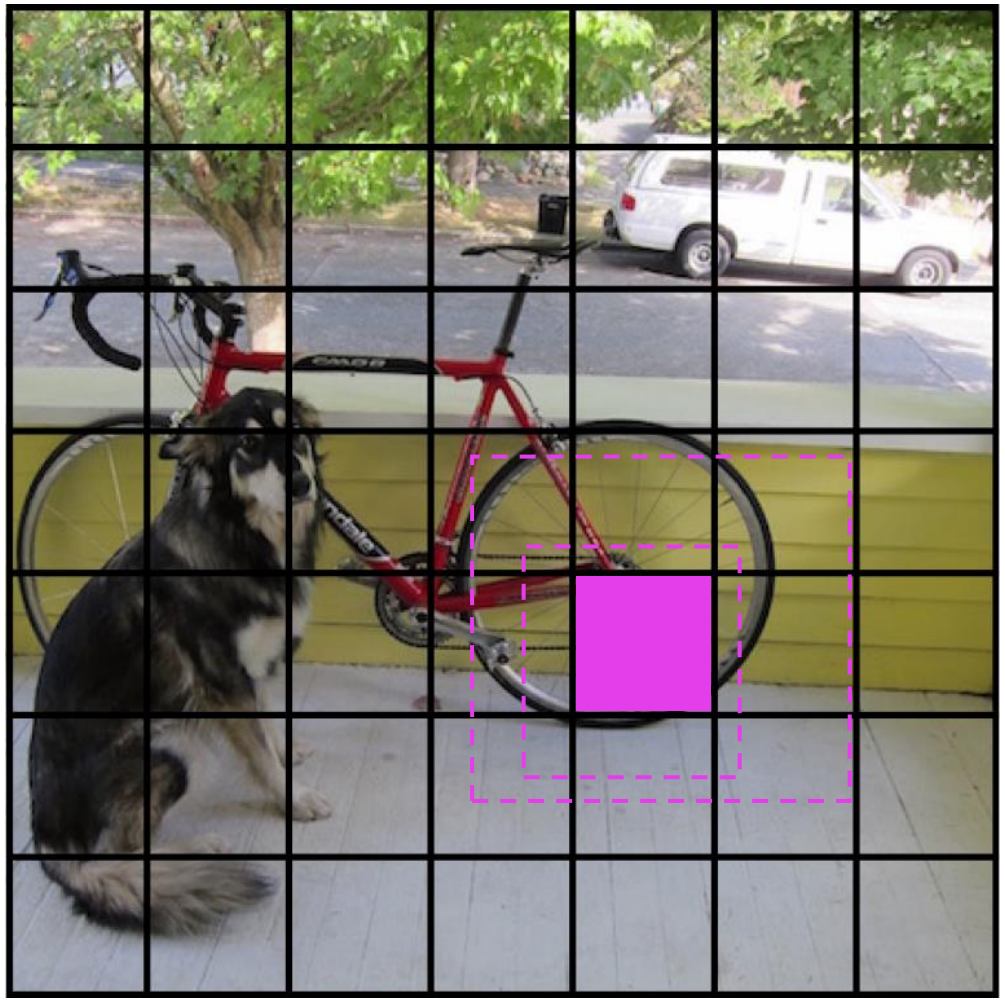
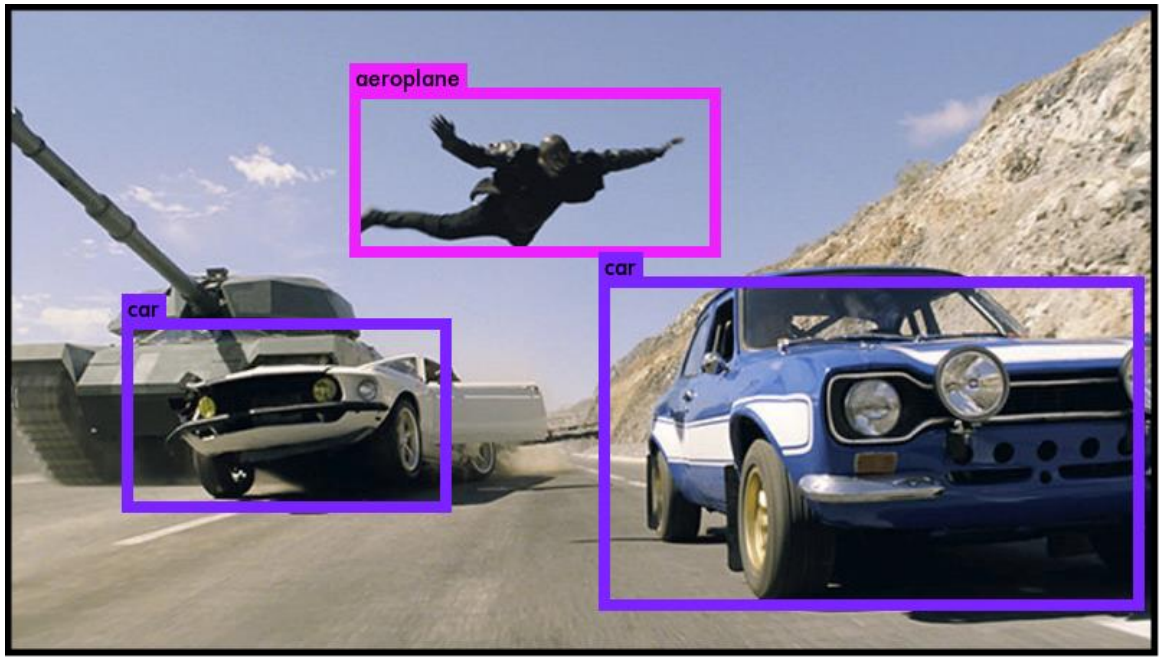


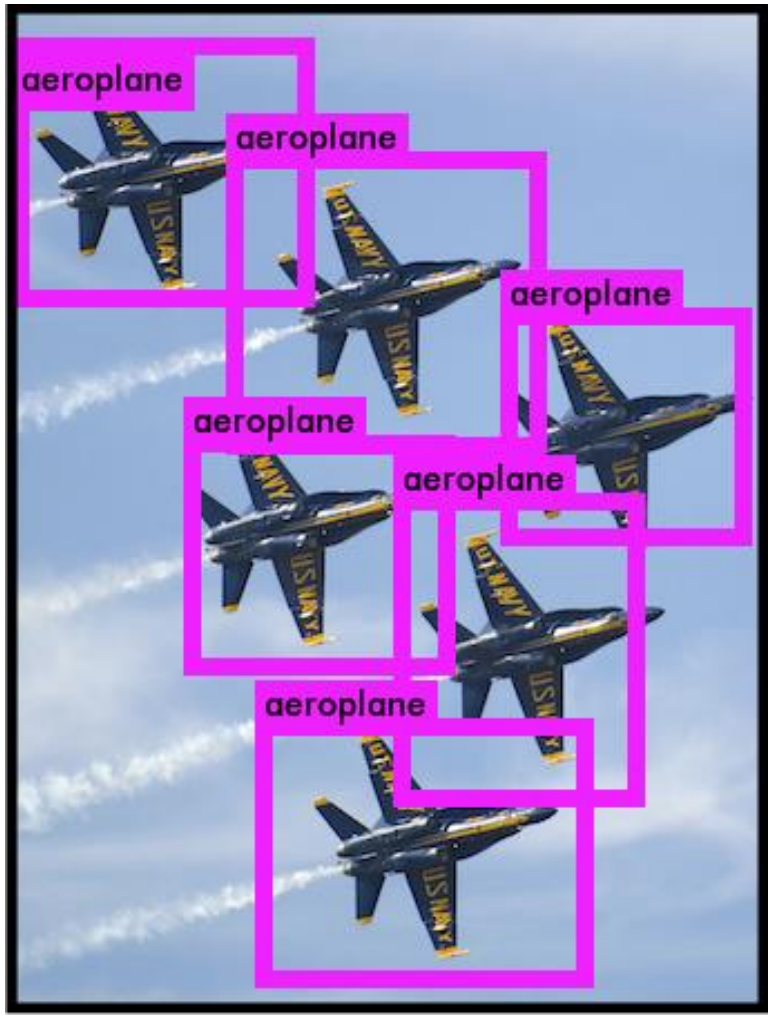
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J. Redmon, S. Divvala, R. Girshick, A. Farhadi. You only look once: Unified, real-time object detection, 2015 (CVPR 2016)

YOLO – Benefits

- Fast. Good for real-time processing
- End-to-end training





J. Redmon, S. Divvala, R. Girshick, A. Farhadi. You only look once: Unified, real-time object detection, 2015 (CVPR 2016)

YOLO – Limitations

- Difficult to detect small objects
- Coarse predictions

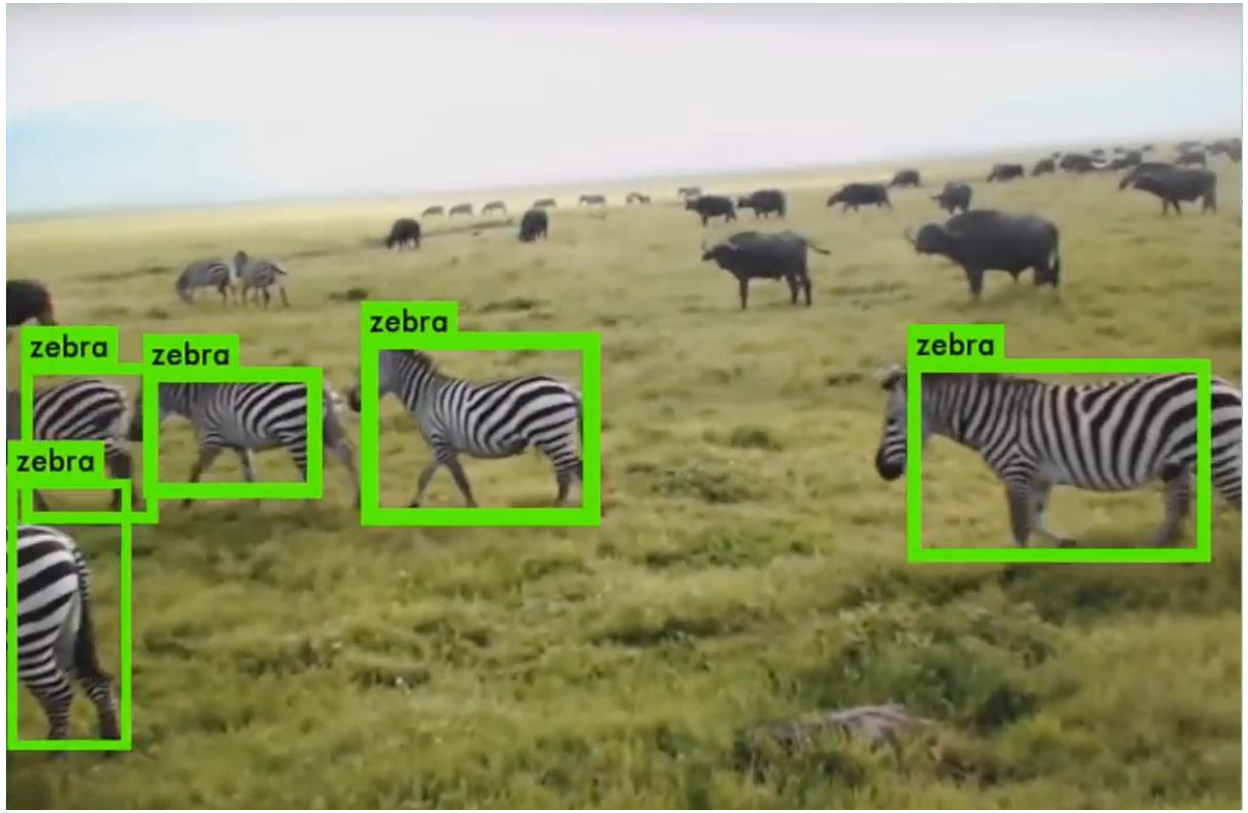
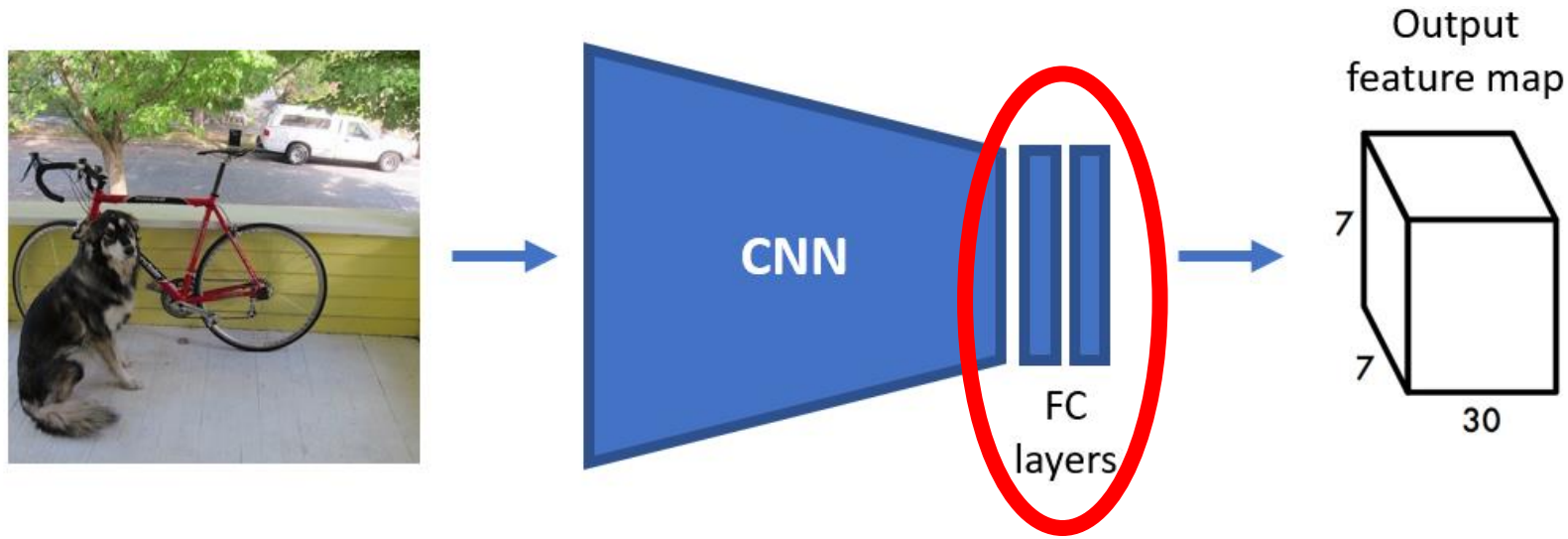


Image credit: https://pjreddie.com/darknet/yolov1/

J. Redmon, S. Divvala, R. Girshick, A. Farhadi. You only look once: Unified, real-time object detection, 2015 (CVPR 2016)

YOLO – Limitations

- Difficult to detect small objects
- Coarse predictions
- Fixed input size



J. Redmon, S. Divvala, R. Girshick, A. Farhadi. You only look once: Unified, real-time object detection, 2015 (CVPR 2016)

YOLO – Limitations

- Difficult to detect small objects
- Coarse predictions
- Fixed input size
- A grid cell can predict only one class

Solutions:

Remove fc layers!

Predict class per bbox (not per cell)

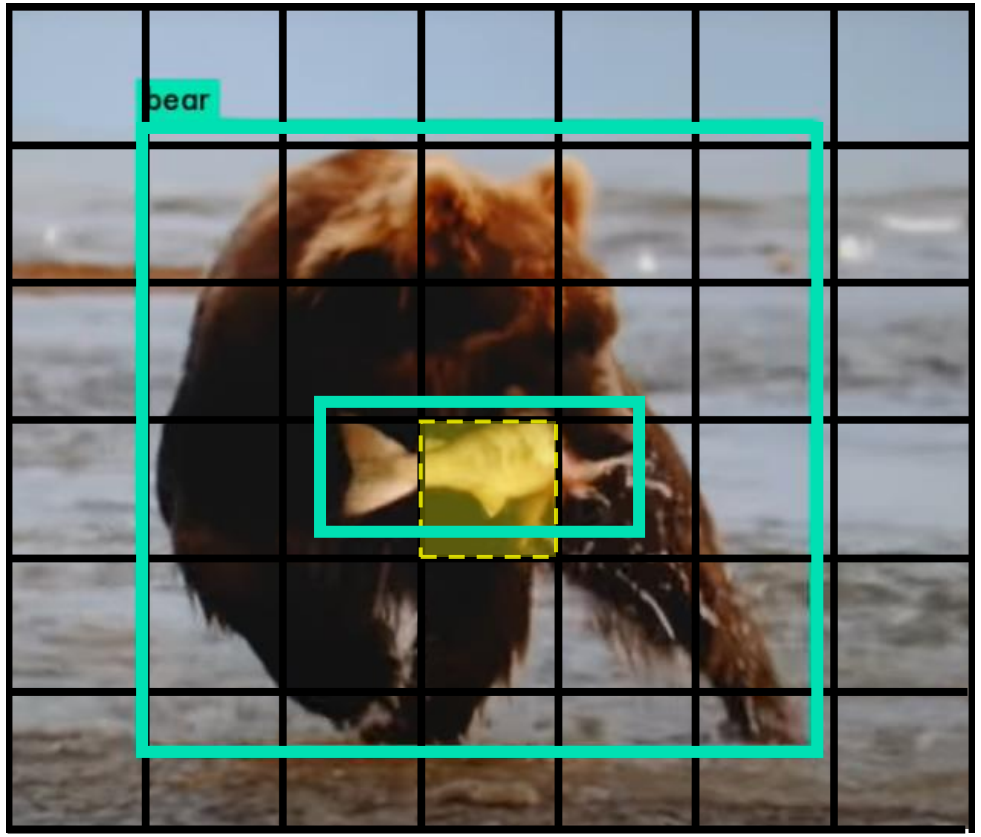
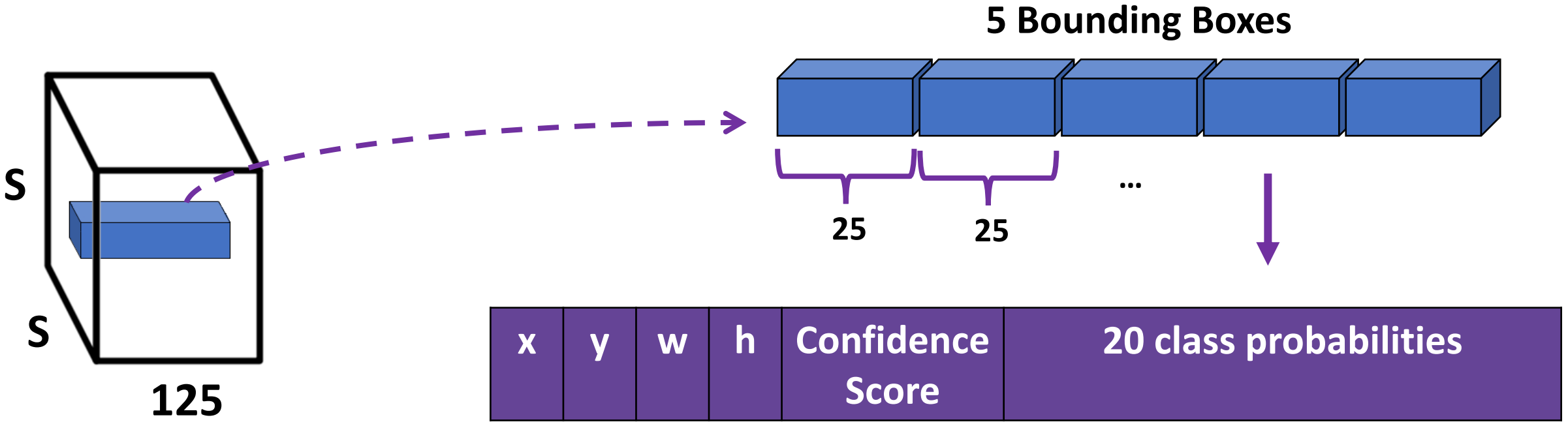


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YOLOv2

- Removed fully connected layers
- A grid cell predicts class probabilities for **each** box



J. Redmon and A. Farhadi. Yolo9000: Better, faster, stronger (CVPR 2017)

YOLOv2

- Removed fully connected layers
- A grid cell predicts class probabilities for **each** box
- Working with anchor boxes (prior bounding boxes)

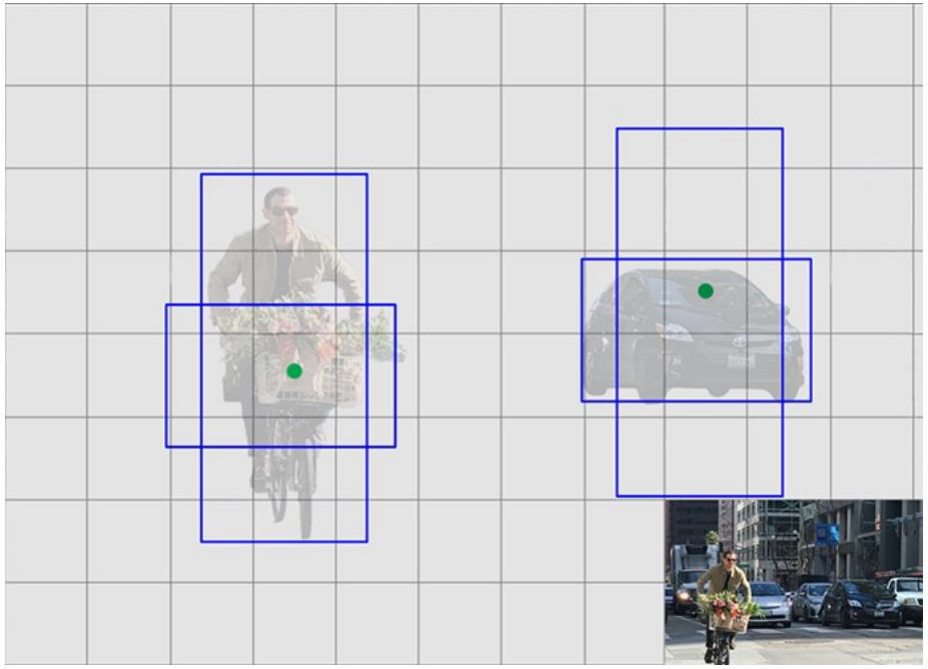
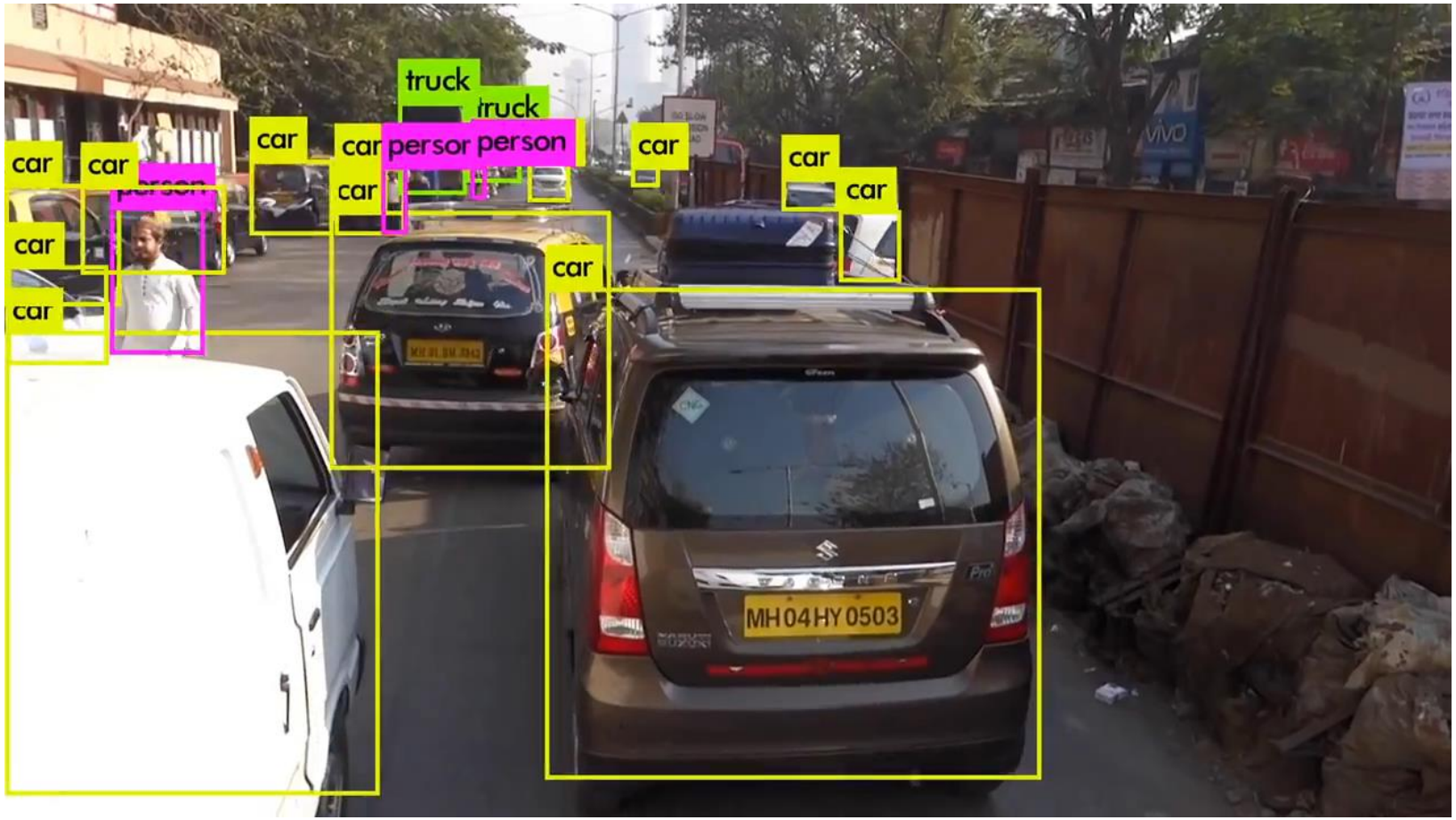


Image credit: medium

J. Redmon and A. Farhadi. Yolo9000: Better, faster, stronger (CVPR 2017)

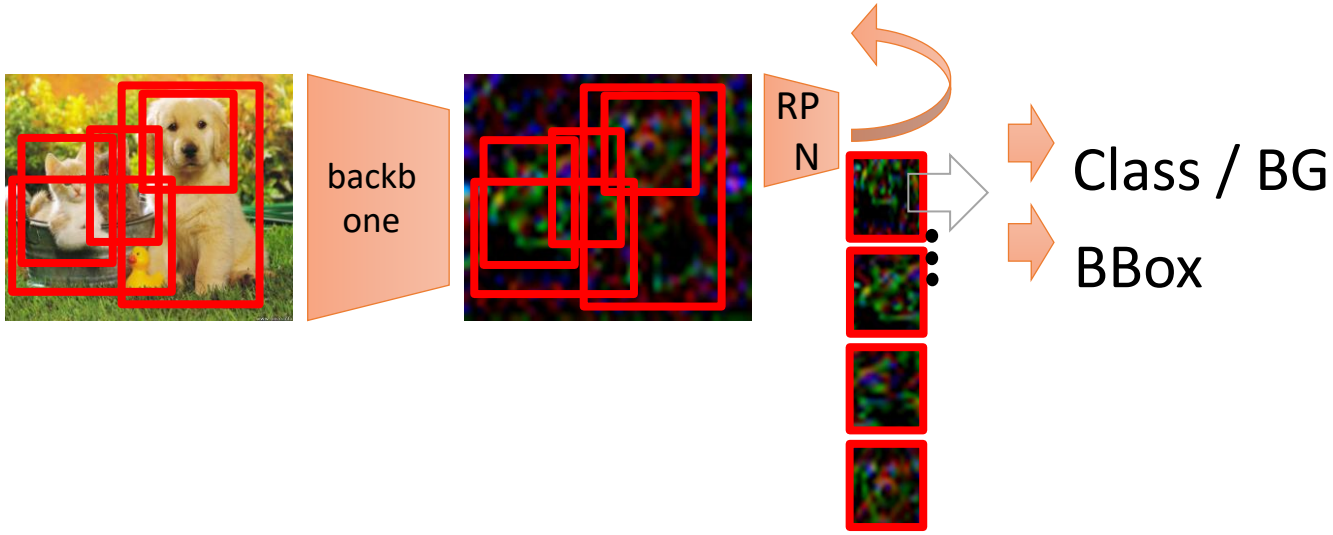
There’s always room for improvement!

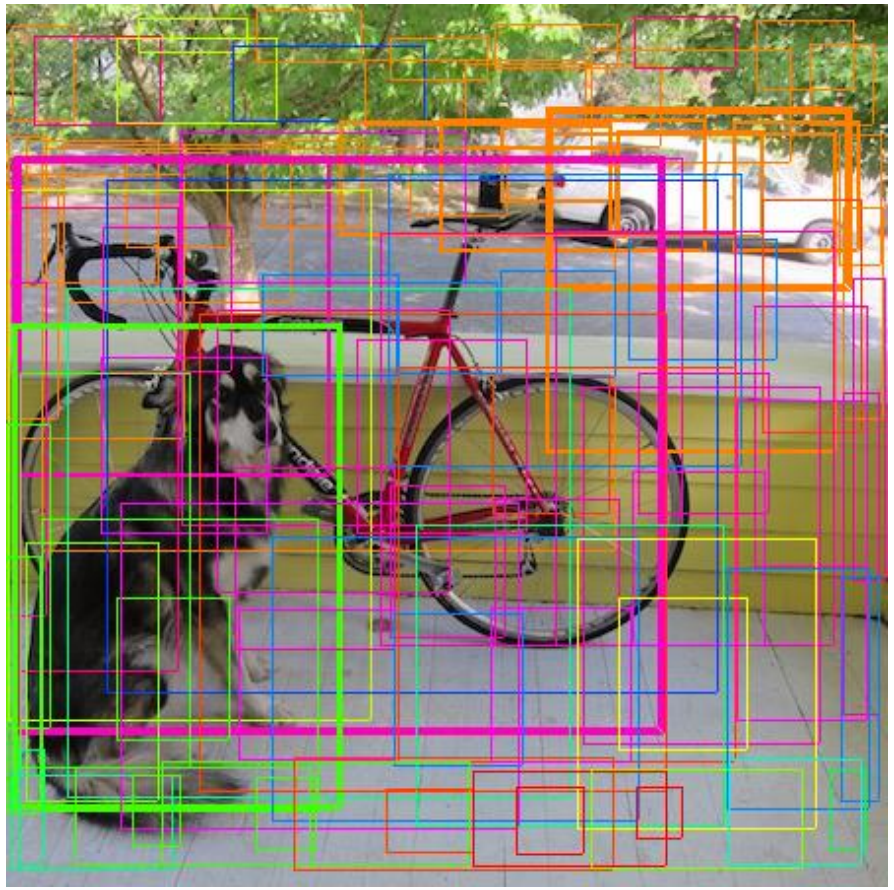
- YOLOv3
  J. Redmon, A. Farhadi. Yolov3: An incremental improvement, 2018
- YOLO v4
  A. Bochkovskiy, C. Wang, H. Liao. Yolov4: Optimal speed and accuracy of object detection (Feb. 2020)
- YOLOv5
  YOLOv5 by ultralytics (June 2020)
- PP-YOLO
  X. Long, K. Deng, G. Wang, Y. Zhang, Q. Dang, Y. Gao, H. Shen, J. Ren, S. Han, E. Ding, S. Wen. Pp-yolo: An effective and efficient implementation of object detector (June 2020)
- PP-YOLOv2 (2021)
  X. Huang, X. Wang, W. Lv, X. Bai, X. Long, K. Deng, Q. Dang, S. Han, Q. Liu, X. Hu, D. Yu, Y. Ma, O. Yoshie. PP-YOLOv2: A Practical Object Detector (2021)
- …



Video: YOLOv3

Object Detection So Far..

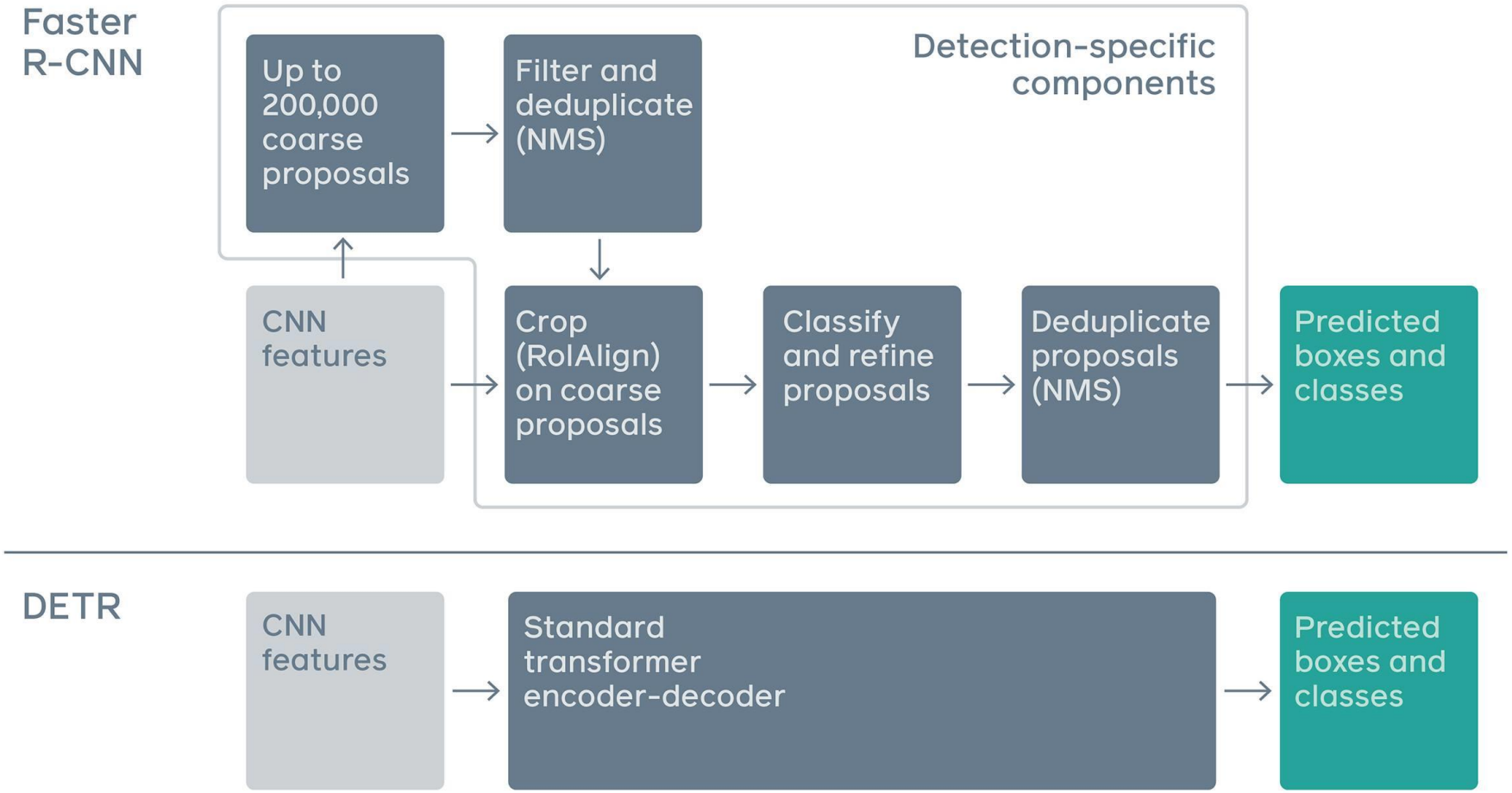




Annoying parts of Detection:
Multiple Bboxs → NMS, Threshold

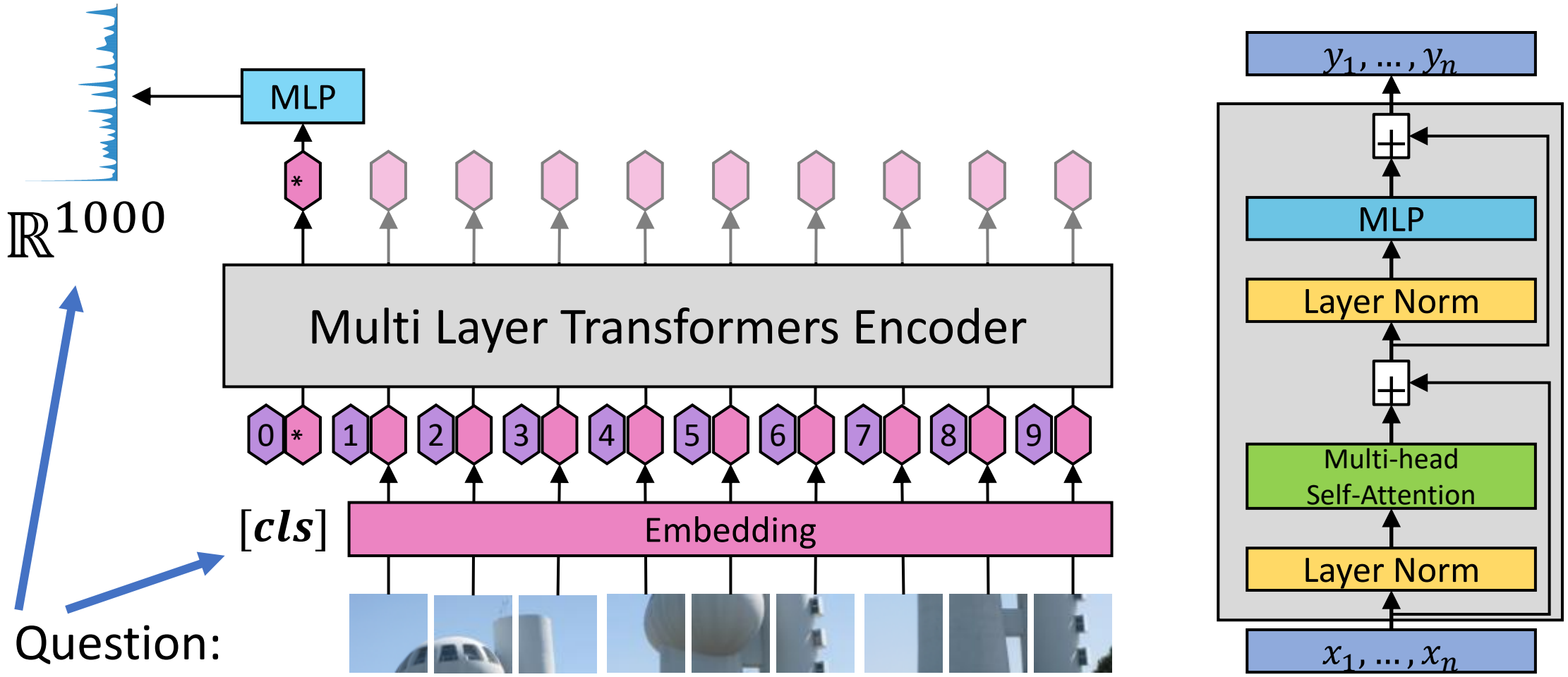
Can we do better? “Set” prediction

DETR: DEtection with TRansformers



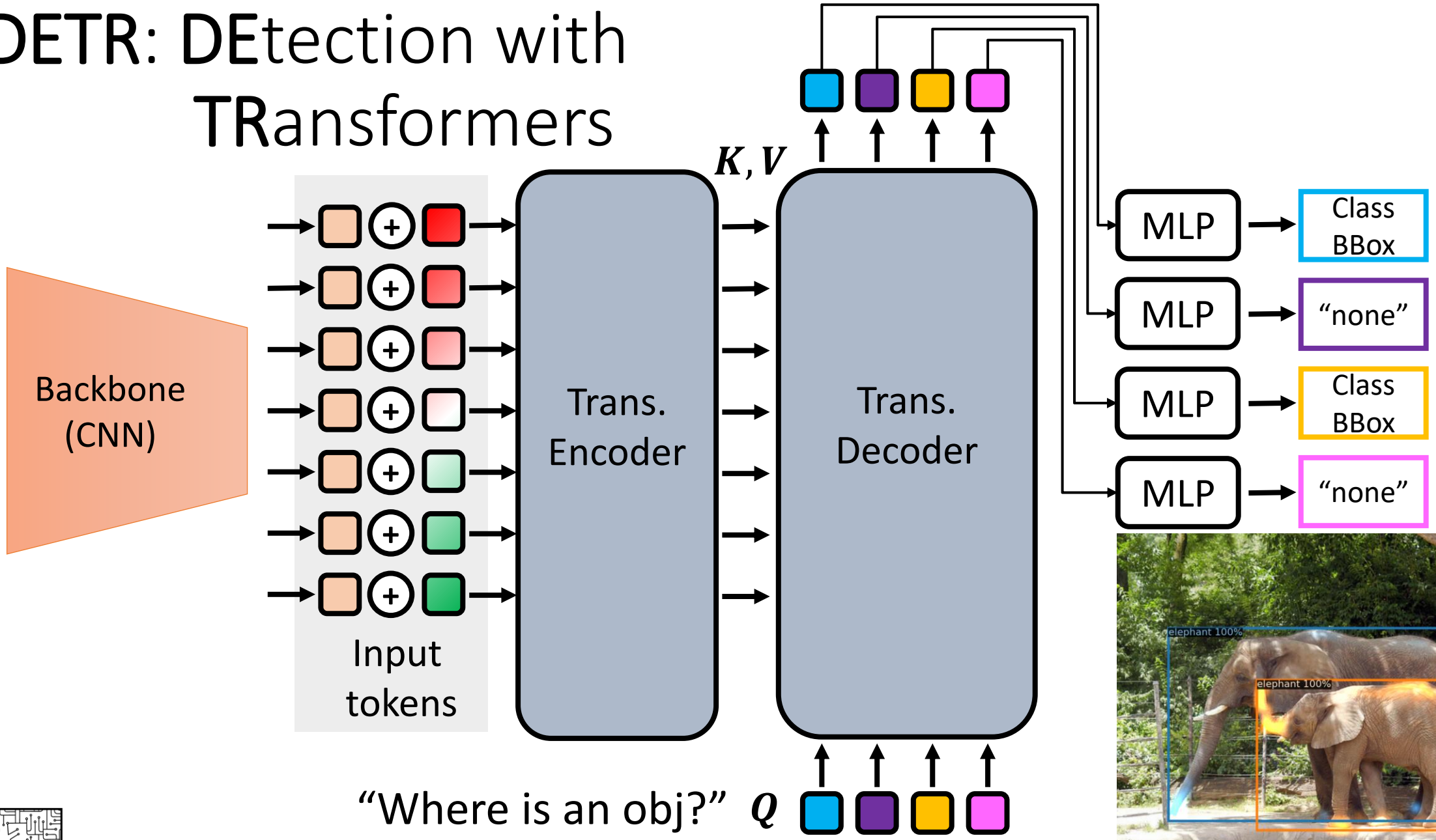
Carion N, Massa F, Synnaeve G, Usunier N, Kirillov A, Zagoruyko S. **DETR: End-to-End Object Detection with Transformers** (ECCV 2020)

Vision Transformers (ViT)



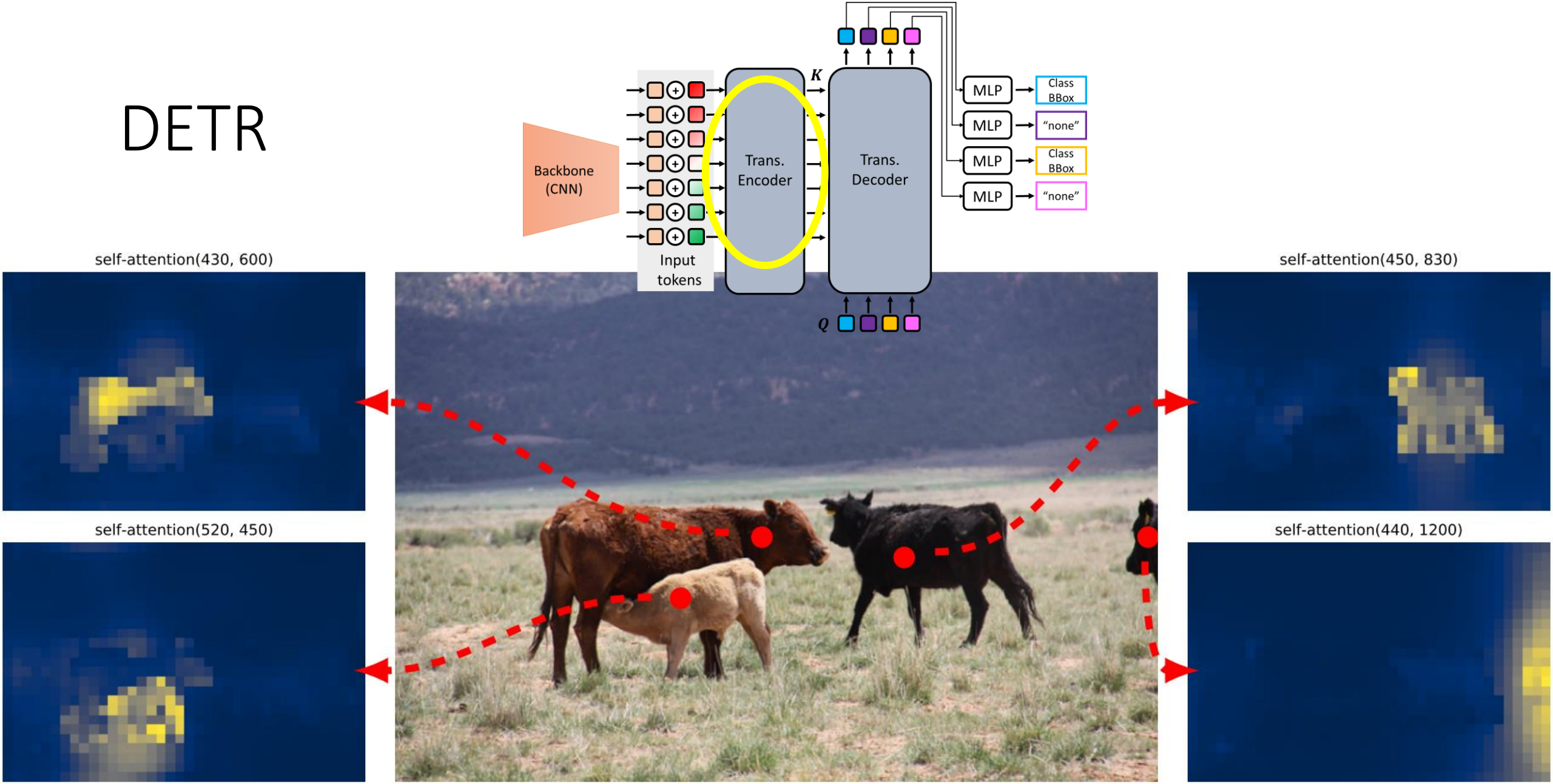
Dosovitskiy A., Beyer L., Kolesnikov A., Weissenborn D., Zhai X., Unterthiner T., Dehghani M., Minderer M., Heigold G., Gelly S., Uszkoreit J. and Houlsby N. "**An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale**" (ICLR 2021)

DETR: DEtection with TRansformers



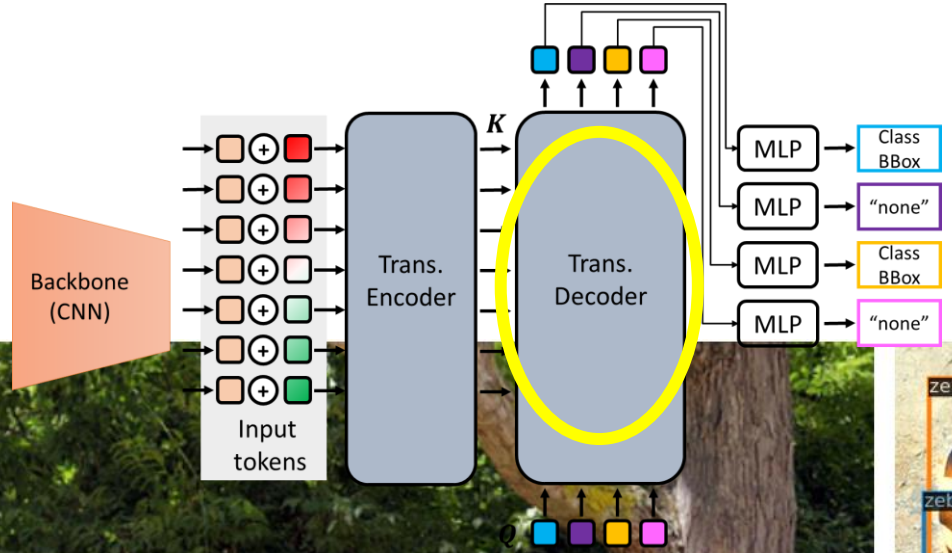
Carion N, Massa F, Synnaeve G, Usunier N, Kirillov A, Zagoruyko S. DETR: End-to-End Object Detection with Transformers (ECCV 2020)

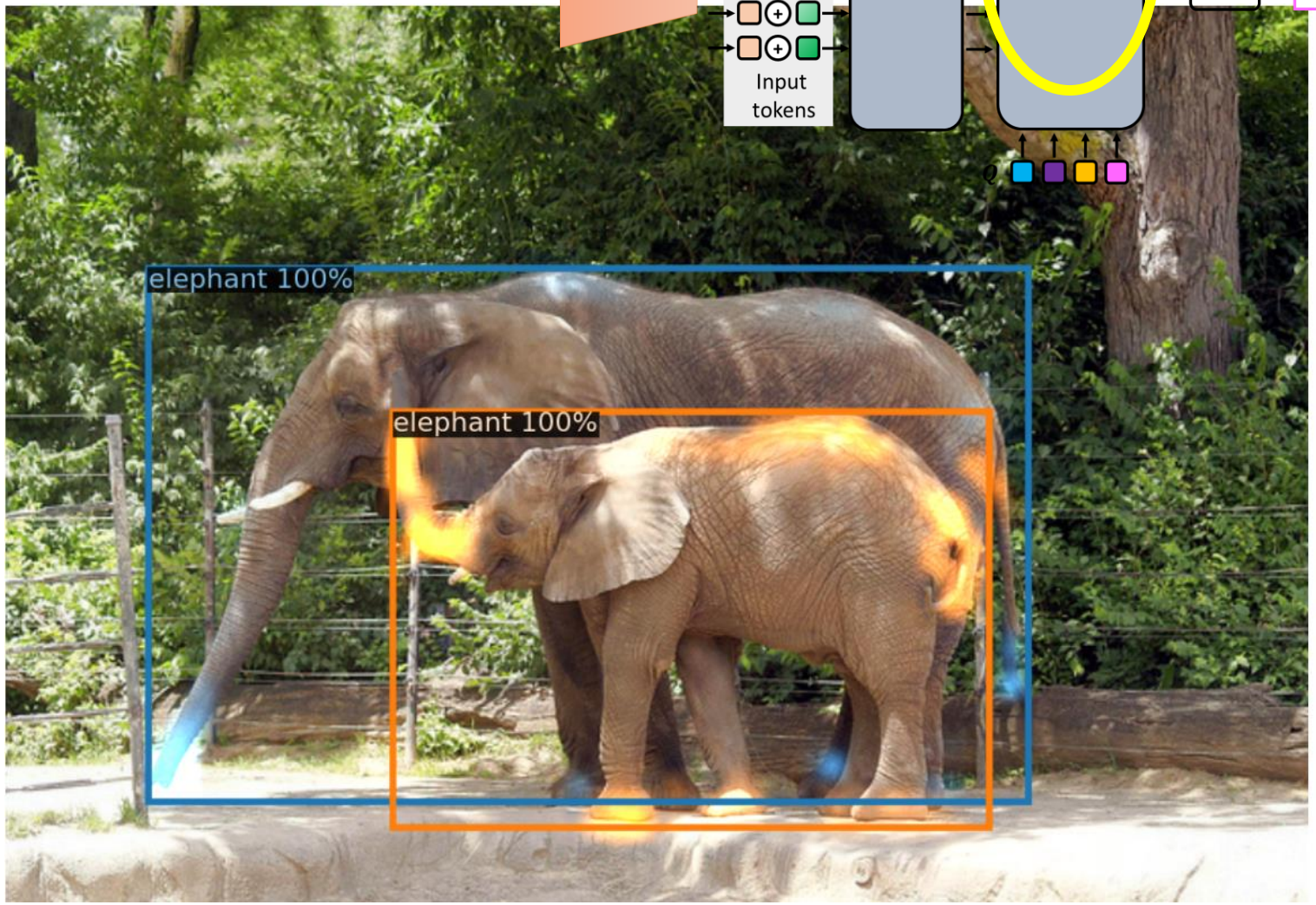
DETR

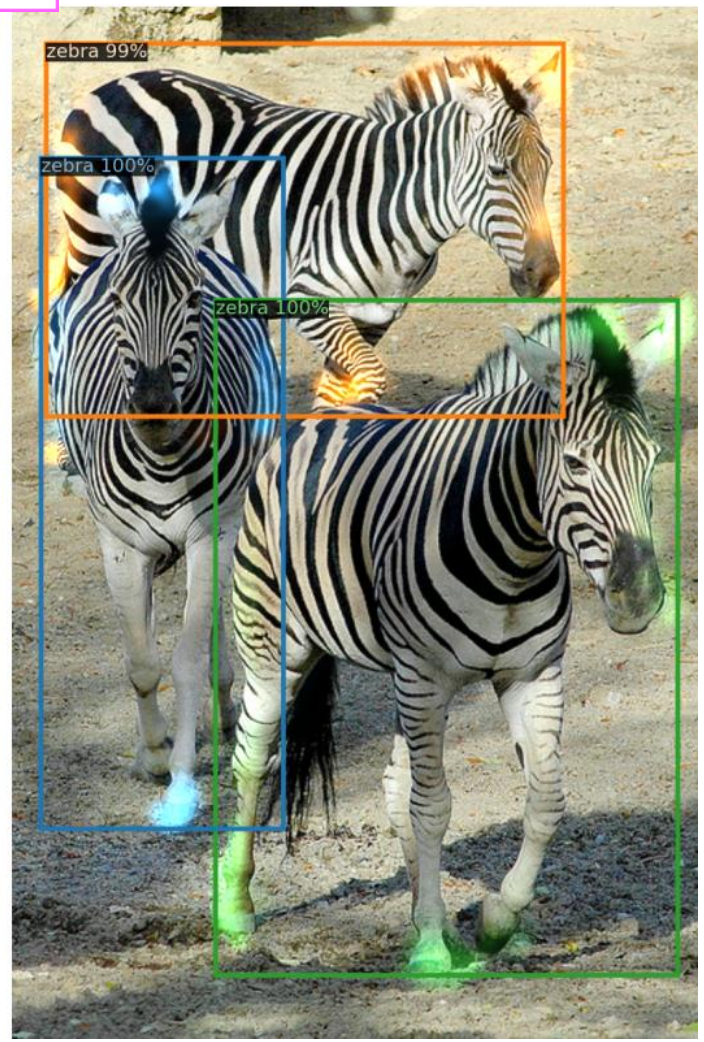


Carion N, Massa F, Synnaeve G, Usunier N, Kirillov A, Zagoruyko S. [DETR: End-to-End Object Detection with Transformers](#) (ECCV 2020)

DETR



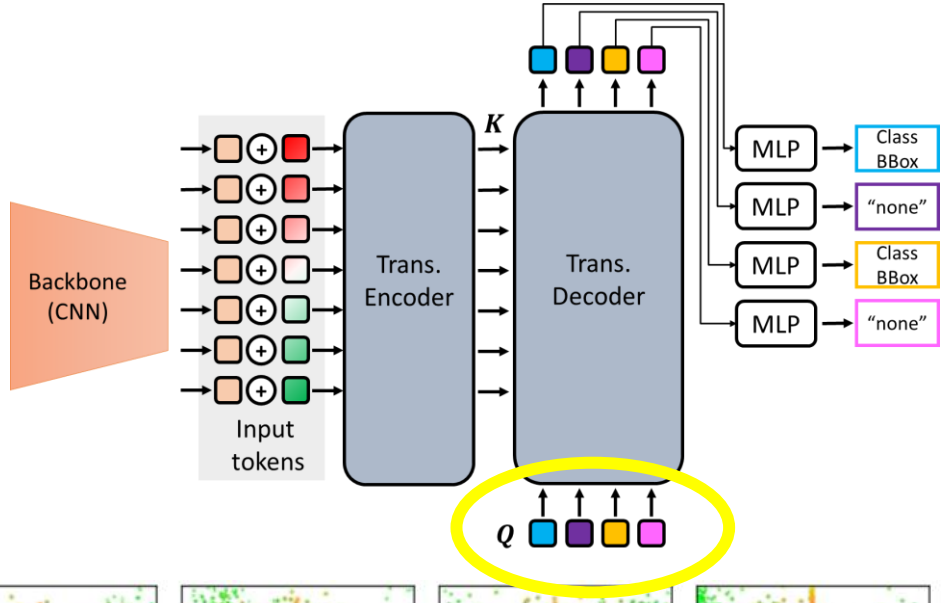


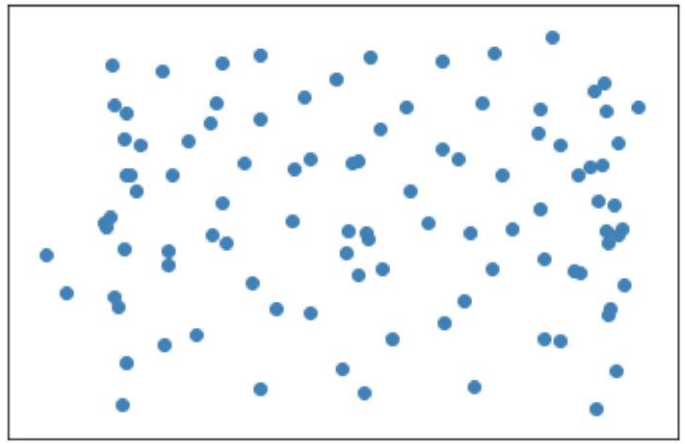


WAIC

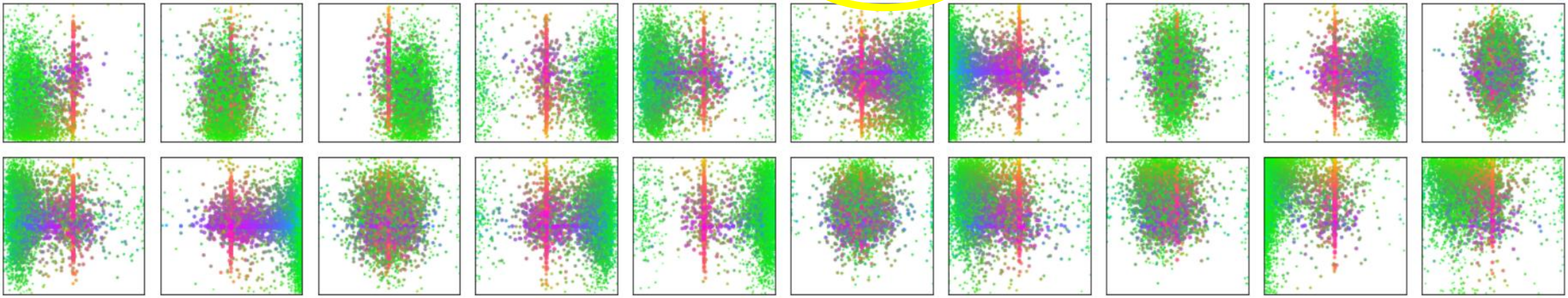
Carion N, Massa F, Synnaeve G, Usunier N, Kirillov A, Zagoruyko S. DETR: End-to-End Object Detection with Transformers (ECCV 2020)

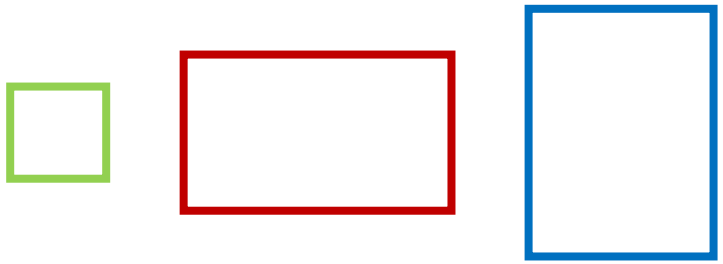
DETR





Average box center predicted by each query





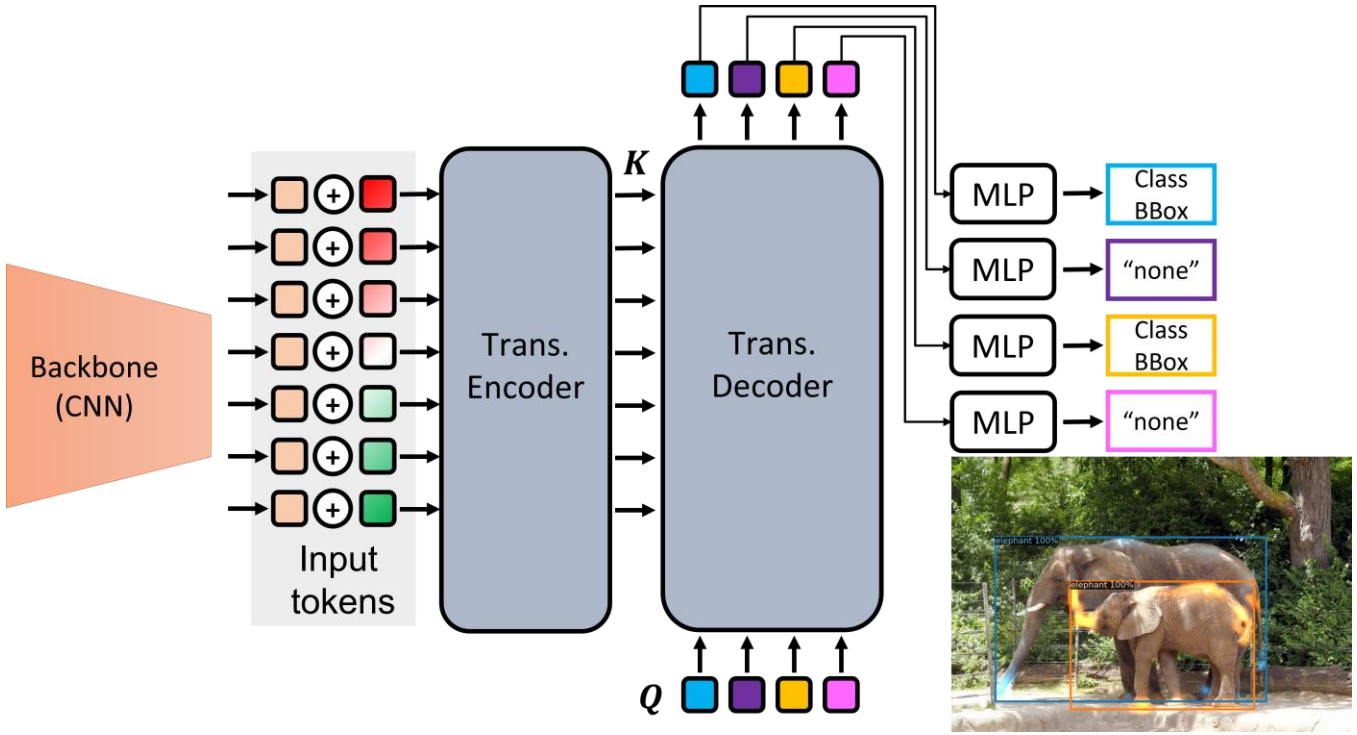
WAIC

Carion N, Massa F, Synnaeve G, Usunier N, Kirillov A, Zagoruyko S. DETR: End-to-End Object Detection with Transformers (ECCV 2020)

DETR

- End-to-end training
- No NMS (!)

- Simple approach
- On par with Faster-R-CNN



Carion N, Massa F, Synnaeve G, Usunier N, Kirillov A, Zagoruyko S. DETR: End-to-End Object Detection with Transformers (ECCV 2020)

Previously

- Object detection
  - Faster RCNN
- Semantic segmentation
  - FCN / DeepLab
  - UNet

Today

- (Multiple) Object detection
  - YOLO (You Only Look Once)
  - DETR (DEtection TRansformer)
- Semantic Segmentation
  - Segmenter

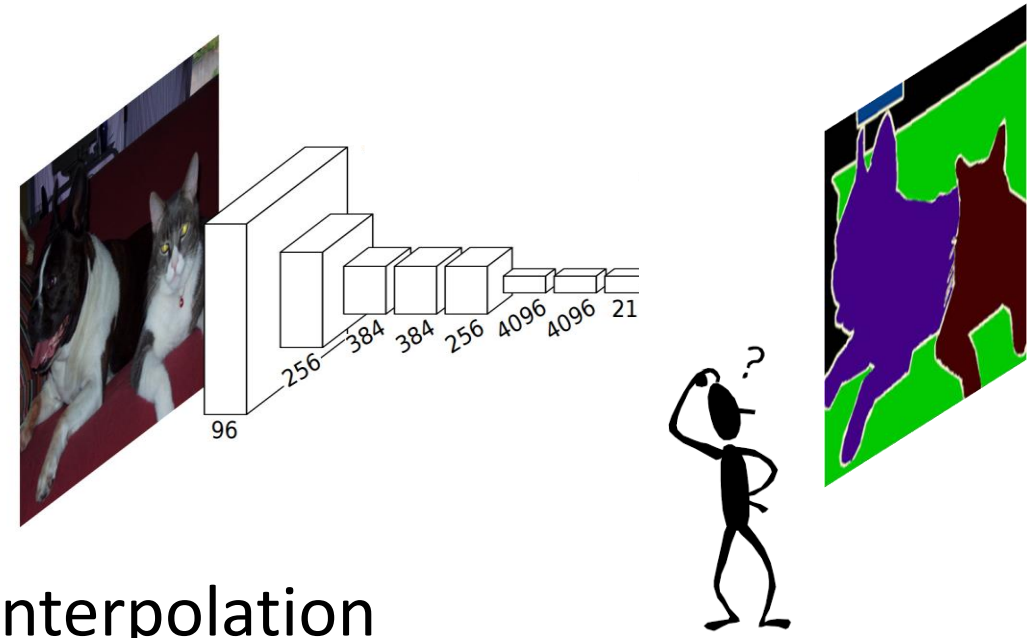
Semantic Segmentation



 Slide credit: Shai Bagon

Semantic Segmentation

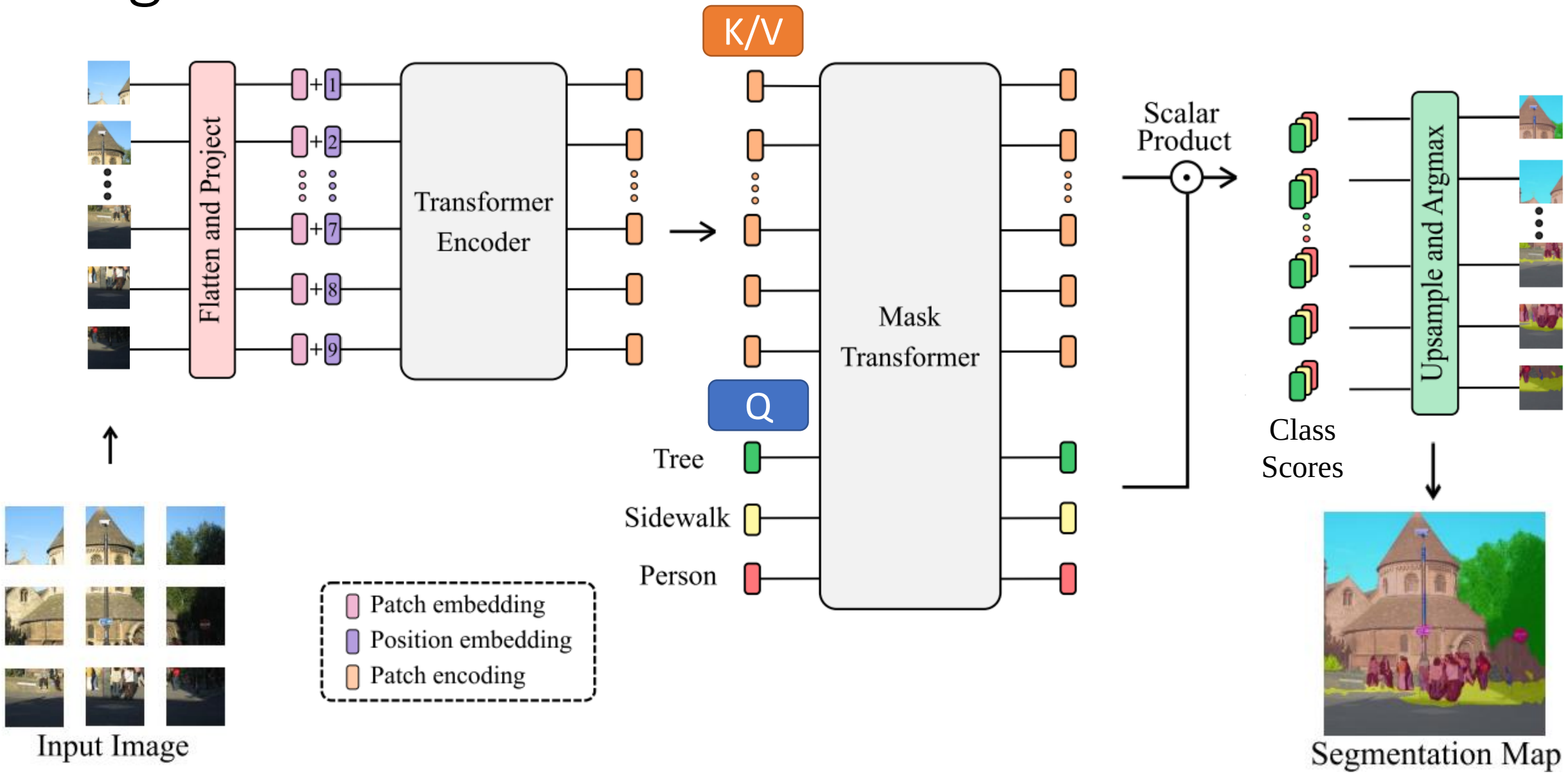
Resolution vs. Semantic information



- FCN: using “transposed convolution”
- DeepLab: dilated convolution + simple interpolation
- U-net: skip connections

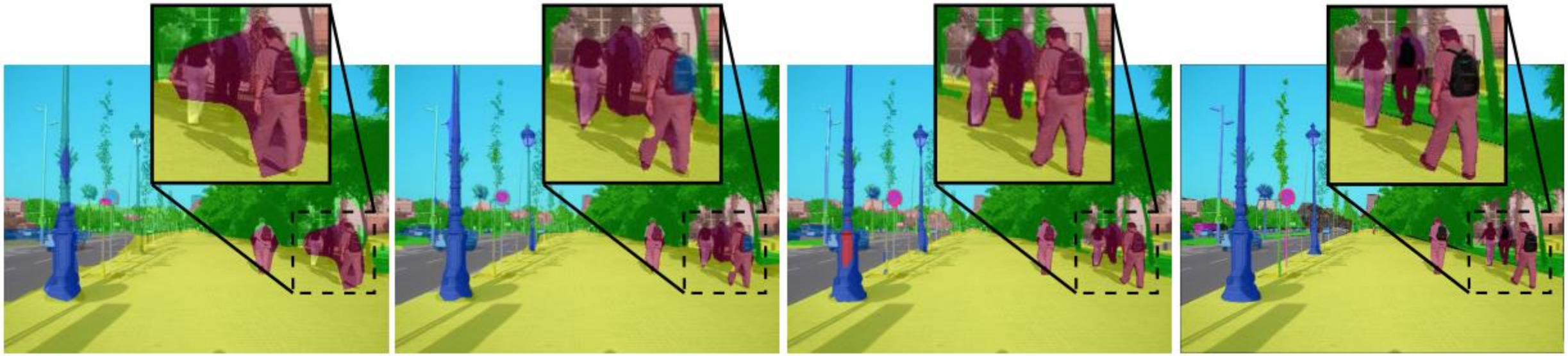
 Slide credit: Shai Bagon

Segmenter



R. Strudel, R. Garcia, I. Laptev, C. Schmid. Segmenter: Transformer for semantic segmentation (ICCV 2021)

Segmenter – Effect of Patch Size



(a) Patch size 32×32 (b) Patch size 16×16 (c) Patch size 8×8 (d) Ground Truth

Fast

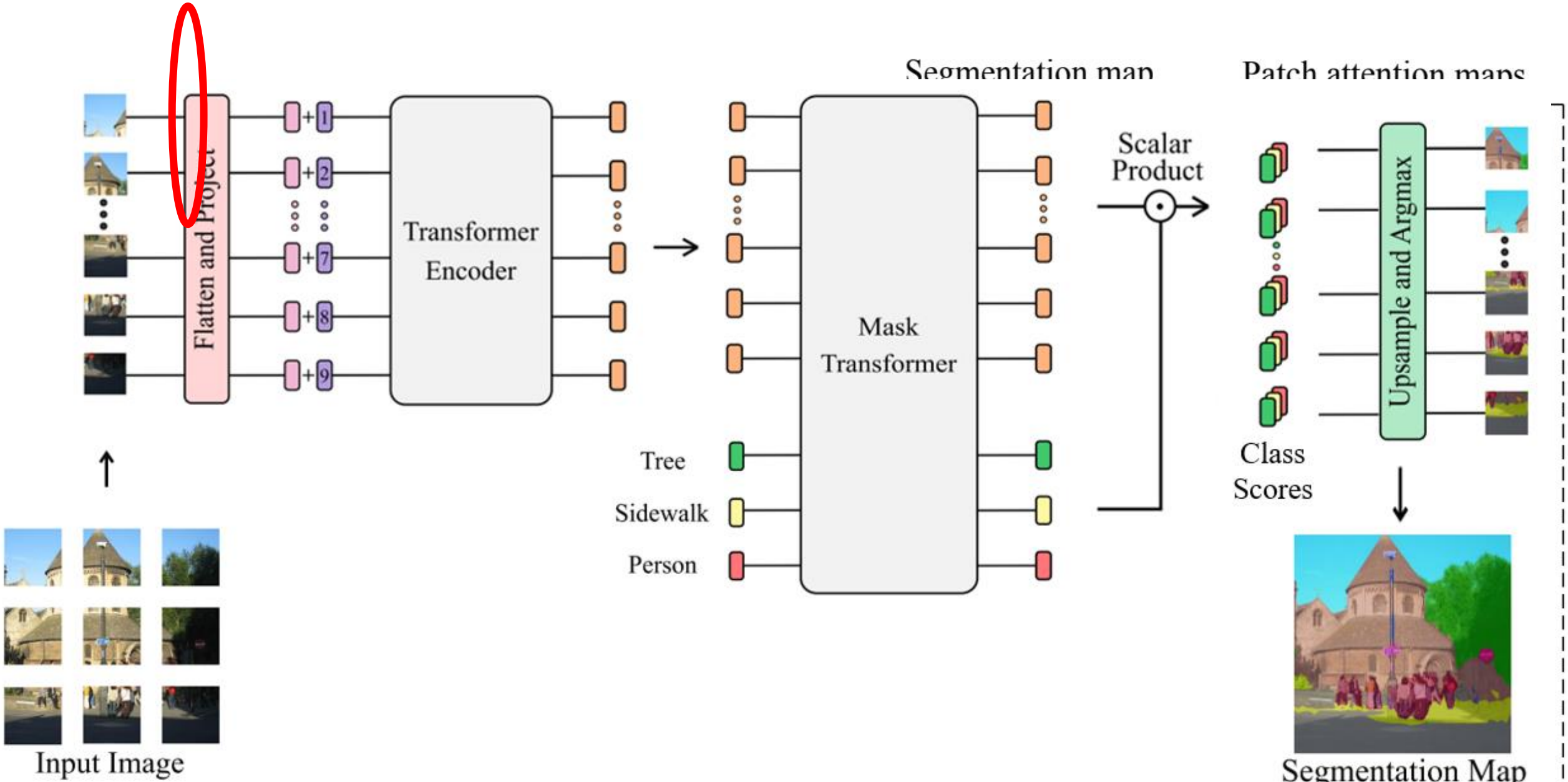
Low accuracy



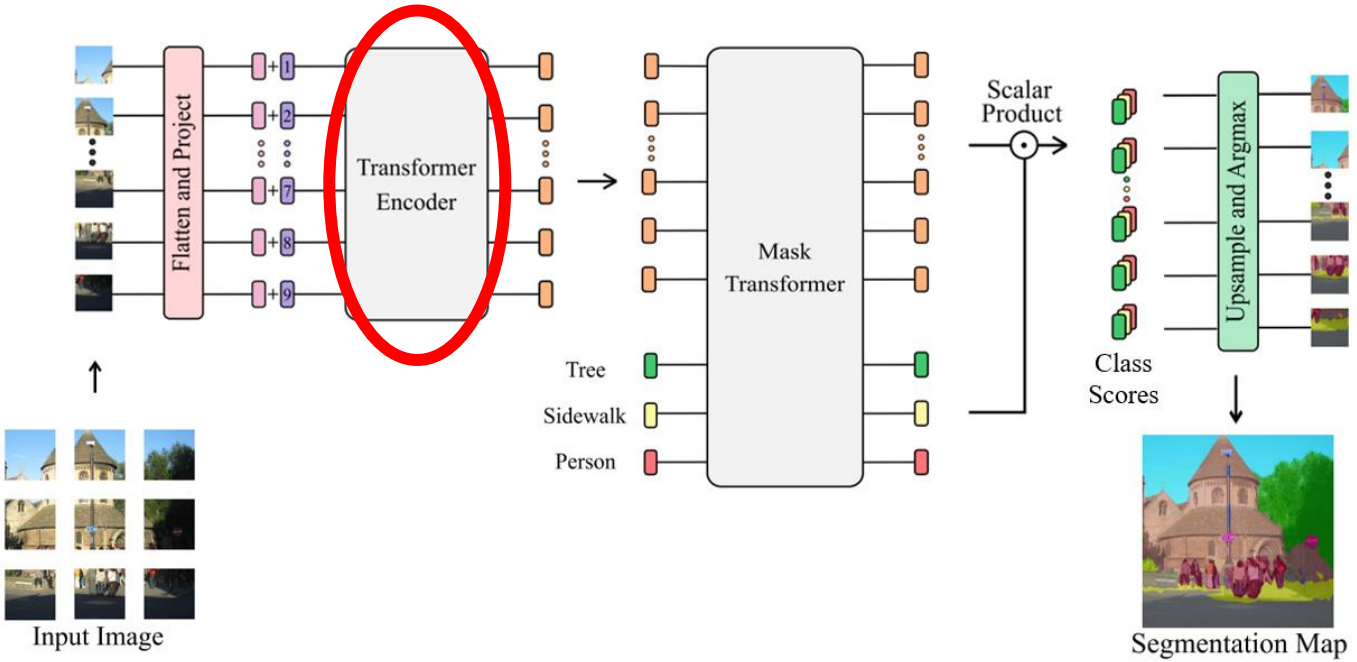
High computational cost

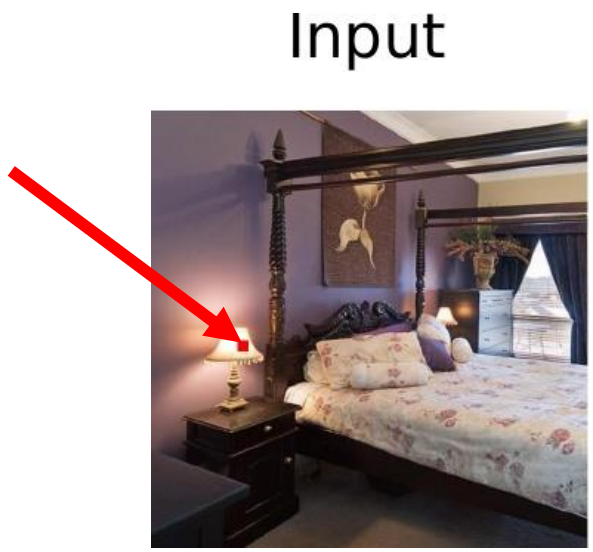
High accuracy

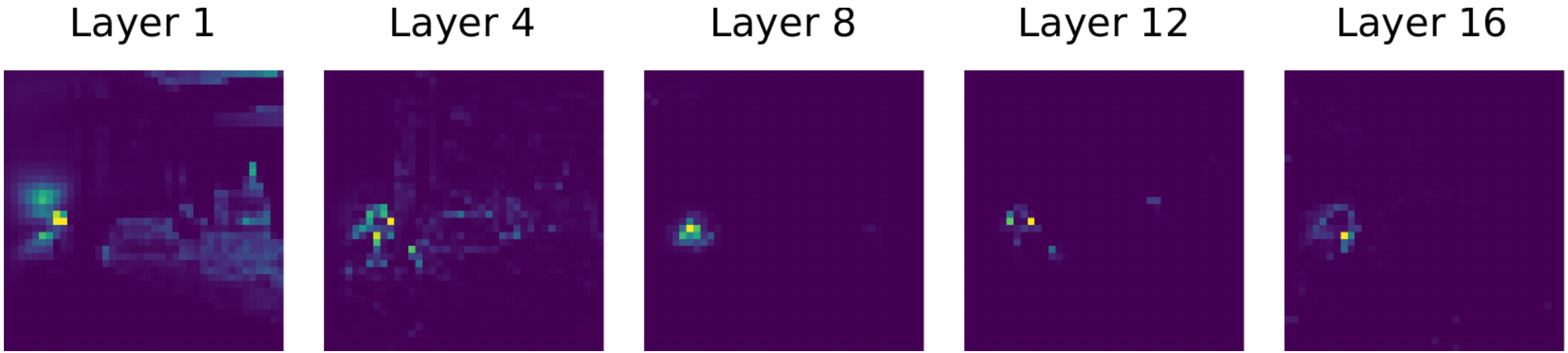
R. Strudel, R. Garcia, I. Laptev, C. Schmid. Segmenter: Transformer for semantic segmentation (ICCV 2021)



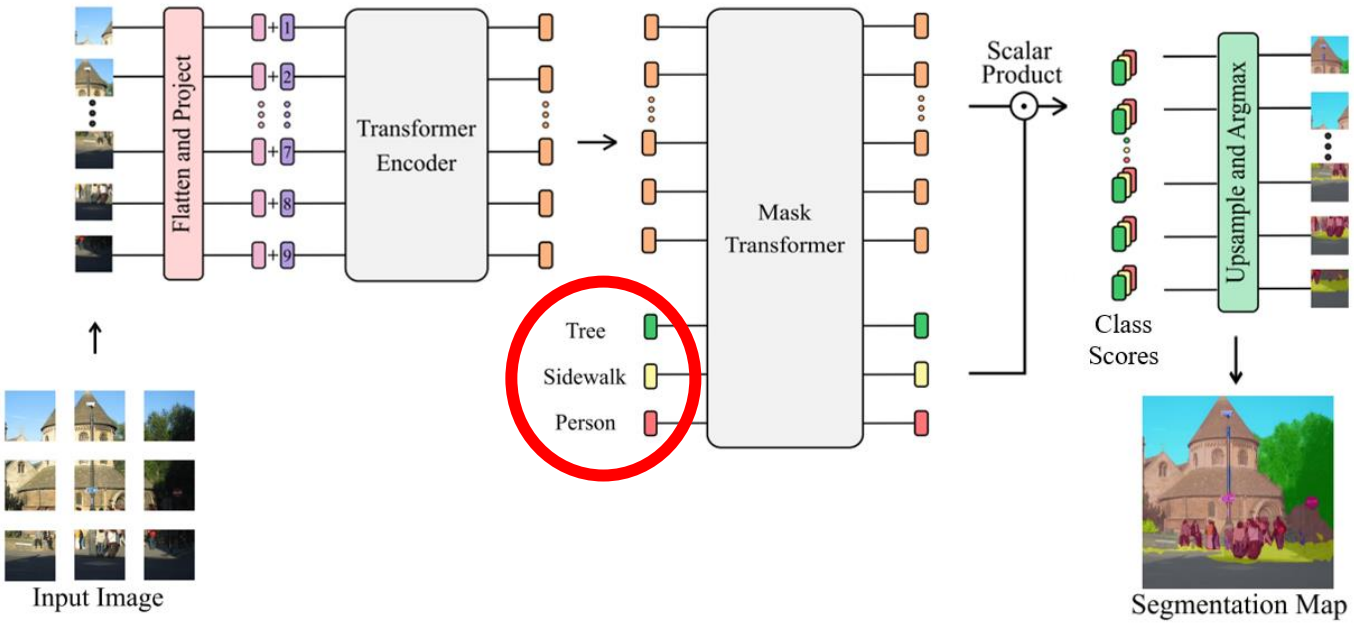
R. Strudel, R. Garcia, I. Laptev, C. Schmid. Segmenter: Transformer for semantic segmentation (ICCV 2021)







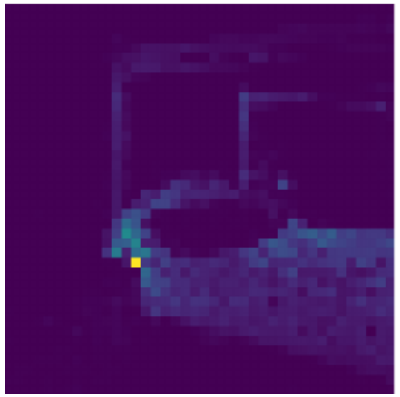
R. Strudel, R. Garcia, I. Laptev, C. Schmid. Segmenter: Transformer for semantic segmentation (ICCV 2021)



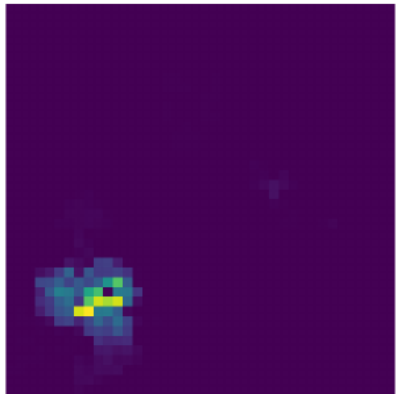
Prediction



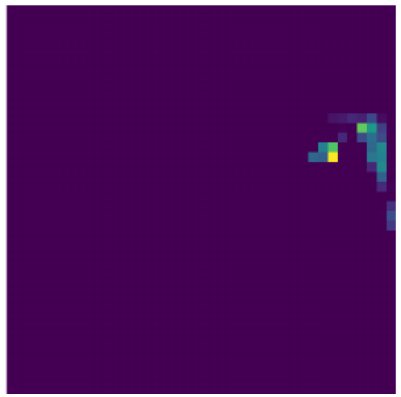
CLS 7 (bed)



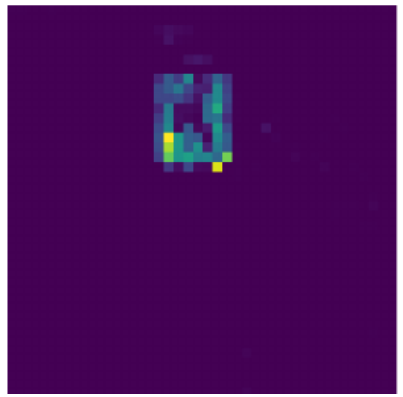
CLS 15 (table)



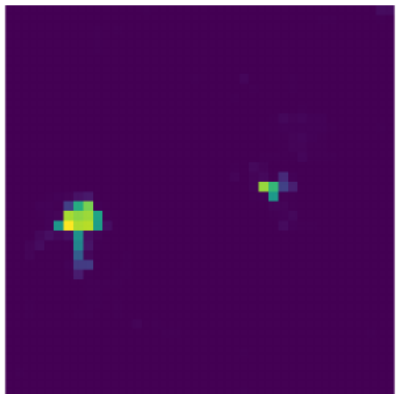
CLS 18 (curtain)



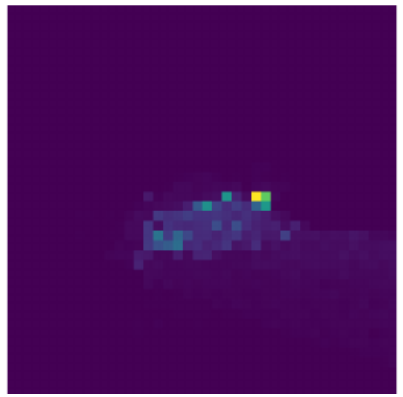
CLS 22 (painting)



CLS 36 (lamp)

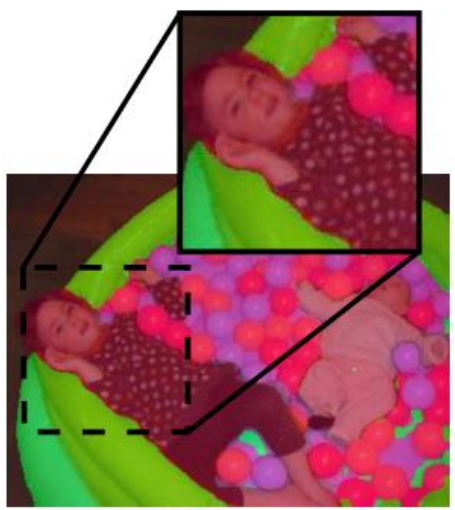


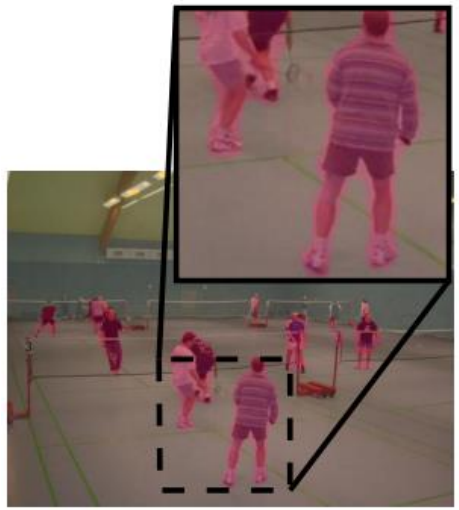
CLS 57 (pillow)

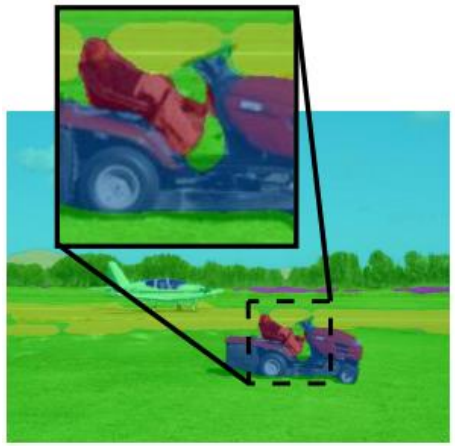


R. Strudel, R. Garcia, I. Laptev, C. Schmid. Segmenter: Transformer for semantic segmentation (ICCV 2021)

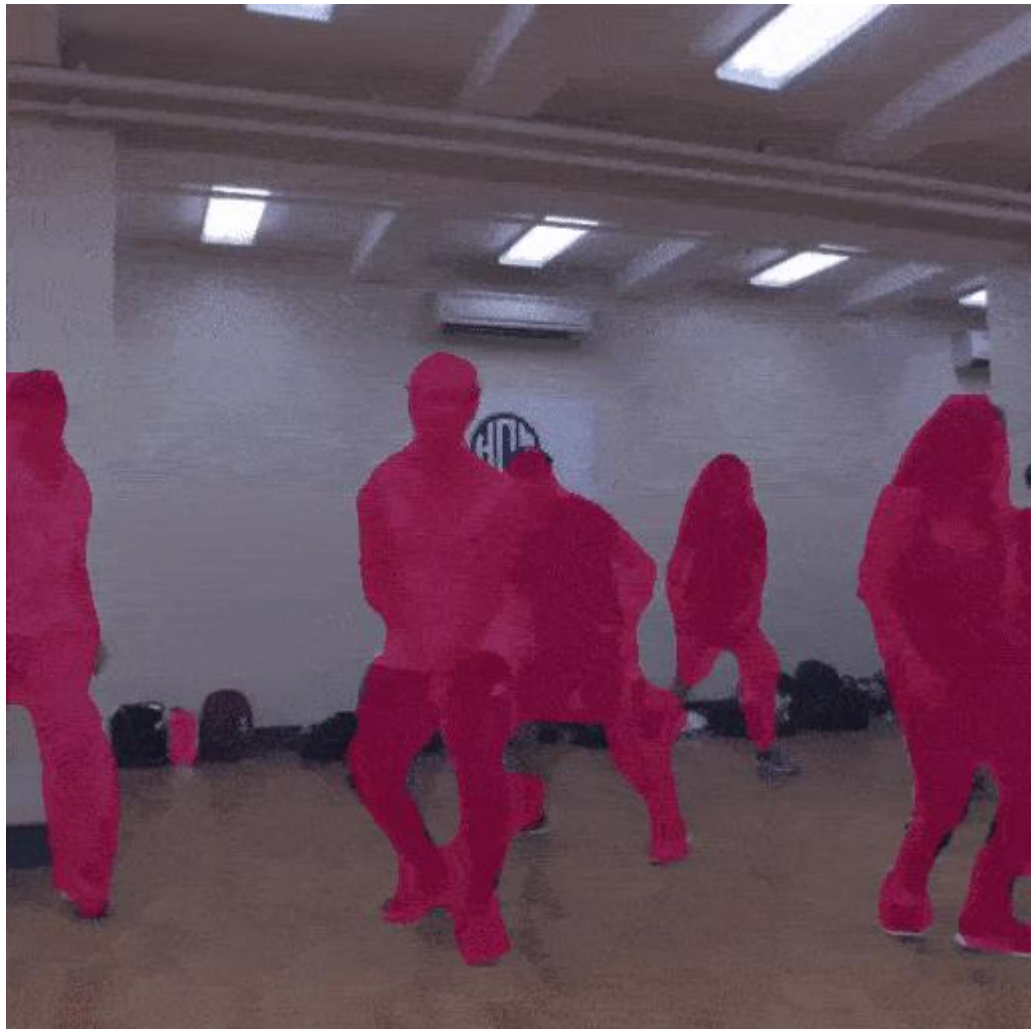
Segmenter











video

R. Strudel, R. Garcia, I. Laptev, C. Schmid. Segmenter: Transformer for semantic segmentation (ICCV 2021)

Today

- (Multiple) Object detection
  - YOLO (You Only Look Once)
  - DETR (DEtection TRansformer)
- Semantic Segmentation
  - Segmenter

Next Week

- Final project expo!